

Surface spectra of Weyl semimetals through self-adjoint extensions

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We apply the method of self-adjoint extensions of Hermitian operators to the low-energy, continuum Hamiltonians of Weyl semimetals in bounded geometries and derive the spectrum of the surface states on the boundary. This allows for the full characterization of boundary conditions and the surface spectra on surfaces both normal to the Weyl node separation as well as parallel to it. We show that the boundary conditions for quadratic bulk dispersions are, in general, specified by a $\mathbb{U}(2)$ matrix relating the wave function and its derivatives normal to the surface. We give a general procedure to obtain the surface spectra from these boundary conditions and derive them in specific cases of bulk dispersion. We consider the role of global symmetries in the boundary conditions and their effect on the surface spectrum. We point out several interesting features of the surface spectra for different choices of boundary conditions, such as a Mexican-hat shaped dispersion on the surface normal to Weyl node separation. We find that the existence of bound states, Fermi arcs, and the shape of their dispersion, depend on the choice of boundary conditions. This illustrates the importance of the physics at and near the boundaries in the general statement of bulk-boundary correspondence.

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I. INTRODUCTION

Topological phases of matter have garnered an increasingly central standing in condensed matter physics and related fields. A generic feature of such phases is an intimate correspondence between nontrivial topology of bulk states, expressed in terms of topological invariants of bulk bands under periodic boundary conditions, and the appearance of robust states bound to the physical edges of the system in a bounded geometry. Examples of this *bulk-boundary correspondence* abound: quantum Hall phases in two dimensions support chiral edge modes [1,2]; time-reversal invariant topological phases in two and three dimensions support gapless edge and surface states [3–8]; topological superconductors support Majorana modes at edges, surfaces, and inside vortex cores [9–12]; and Weyl semimetals are known to support unconventional surface states with open “Fermi arcs” [13–15]. Intuitively, the appearance of these robust surface states regardless of local details of the surface is the result of the global, topological character of the bulk.

Despite many examples and its intuitive appeal, a precise theory of the bulk-boundary correspondence is lacking [16–18]. Surface states are, of course, not special to topological phases. Tamm [19] and then Shockley [20] showed a long time ago that terminating a periodic potential generically leads to the appearance of states bound to the termination surface, whose energy and wave function depend on the boundary conditions at the surface. The role of boundary conditions for surface states of topological phases is, however, not systematically studied. The form of the boundary condition is often not specified or assumed ad hoc [21,22]. In many cases, the bulk Hamiltonian is stripped down to a linear bulk dispersion [23], which would then only support a restricted set of boundary conditions, say, characterized by a single parameter [24,25], which artificially

limits the range of possible physical effects at the surface. A recent work [26] considers a more general dispersion; however, as our approach illustrates, the conditions imposed for the boundary conditions in this work are also too restrictive; as a result, not all physically possible boundary conditions are accounted for.

In this work, we address this question by studying the effect of boundary conditions on surface spectra of Weyl semimetals in a bounded geometry in the most general terms. We do so by studying the generic low-energy, continuum Hamiltonians of a Weyl semimetal and their self-adjoint extensions in bounded geometries [27–29]. This approach allows us to find all physically possible boundary conditions for the given continuum Hamiltonian. Further restrictions on the boundary conditions can be imposed by symmetries of the bulk Hamiltonian. However, it must be emphasized that respecting those symmetries is not necessary at the boundary, where bulk symmetries may be broken extraneously or spontaneously [30]. While not every boundary condition is easily realized or practical in the laboratory, each boundary condition represents physics that is, in principle, possible at and near the boundary.

Specifically, we consider two generic Hamiltonians supporting a pair of degenerate Weyl nodes: one with a minimal bulk dispersion, and the other with an isotropic bulk dispersion. We show that the boundary conditions are in general specified by a $\mathbb{U}(2)$ matrix relating the wave function and its derivatives normal to the surface. On a surface parallel to the Weyl node separation, we find conditions for the existence of Fermi arcs. In particular, for the minimal dispersion, we fully characterize these conditions in terms of a $\mathbb{U}(1)$ phase. For the isotropic dispersion, we find the subset of $\mathbb{U}(2)$ boundary conditions that support straight-line Fermi arcs connecting the Weyl nodes on the surface. In both cases, all self-adjoint boundary

conditions spontaneously break particle-hole symmetry on these surfaces. Moreover, we show that with the simplest boundary condition where the wave function vanishes on the surface, the surface spectrum is linear, with surface bound states whose confinement length is independent of the dispersion and a standing-wave modulation normal to the surface. Moreover, for the surface normal to the Weyl node separation, we characterize the surface bound state spectrum fully in the case of the minimal dispersion and give specific results for the isotropic dispersion. For the minimal dispersion, the energy bands are those of a gapped Dirac spectrum. Interestingly, for the minimal dispersion, the surface states only exist over a range of momenta determined by spin-orbit coupling, and their energy bands are gapped in the shape of a Mexican hat. This raises the possibility of realizing exotic electronic phases at the surface in the presence of interactions or by local potential engineering.

II. MODEL

For a Weyl semimetal with nodes separated by a wave vector k_0 much smaller than the inverse lattice spacing, we can take a minimal continuum model with Bloch Hamiltonian

$$H_{\mathbf{k}} = \begin{bmatrix} \epsilon_{\mathbf{k}} & \lambda(k_x - ik_y) \\ \lambda(k_x + ik_y) & -\epsilon_{\mathbf{k}} \end{bmatrix} \equiv \mathbf{d}_{\mathbf{k}} \cdot \boldsymbol{\sigma}, \quad (1)$$

where $\mathbf{k} = (k_x, k_y, k_z)$ is the crystal momentum (we use natural units $\hbar = 1$), $\boldsymbol{\sigma}$ is the vector of Pauli matrices, $\mathbf{d}_{\mathbf{k}} = (\lambda k_x, \lambda k_y, \epsilon_{\mathbf{k}})$, λ parametrizes the spin-orbit interaction, and $\epsilon_{\mathbf{k}}$ is the bulk dispersion relation in the absence of spin-orbit coupling. We shall consider two dispersion relations: (1) $\epsilon_{\mathbf{k}} = (k_z^2 - k_0^2)/2m$ corresponds to the minimal model of Ref. [22]; and (2) $\epsilon_{\mathbf{k}} = (k^2 - k_0^2)/2m$ is isotropic in $\mathbf{k}$. Here $k_0, m > 0$ are constants.

The bulk spectrum is given by $\pm E_{\mathbf{k}}$,

$$E_{\mathbf{k}} = |\mathbf{d}_{\mathbf{k}}| = \sqrt{\epsilon_{\mathbf{k}}^2 + \lambda^2(k_x^2 + k_y^2)}. \quad (2)$$

There are two Weyl nodes at $\pm \mathbf{k}_0$ with $\mathbf{k}_0 = (0, 0, k_0)$. The winding of the vector $\mathbf{d}_{\mathbf{k}}$ in the Brillouin zone determines the existence of the Weyl nodes through the monopoles of the Berry's flux. The monopole charge is given by

$$\text{sgn det} \frac{\partial \mathbf{d}_{\mathbf{k}}}{\partial \mathbf{k}} \bigg|_{\mathbf{k}=\pm \mathbf{k}_0} = \pm 1. \quad (3)$$

In the following, we will work with dimensionless quantities $\mathbf{k}/k_0 \rightarrow \mathbf{k}$, $2m\lambda/k_0 \rightarrow \lambda$, and $2mE/k_0^2 \rightarrow E$. Then the two dispersions take the form (1) $\epsilon_{\mathbf{k}} = k_z^2 - 1$ and (2) $\epsilon_{\mathbf{k}} = k^2 - 1$.

We now note some of the symmetries of the bulk model:

(1) There is an inversion symmetry represented by the unitary $\Pi = \sigma_z$:

$$\Pi H_{\mathbf{k}} \Pi = H_{-\mathbf{k}}. \quad (4)$$

We note that this is not the same as the parity in the full system, which is broken in the Weyl semimetal. This symmetry describes the symmetric splitting of Weyl nodes in the minimal model and is therefore only approximate in a local neighborhood around the pair of Weyl nodes in the Brillouin zone.

(2) There is an antiunitary particle-hole symmetry represented by $\Gamma = \sigma_y K$, $\Gamma^2 = -\mathbb{1}$, where K is the complex conjugation:

$$\Gamma H_{\mathbf{k}} \Gamma^{-1} = \sigma_y H_{\mathbf{k}}^* \sigma_y = -H_{\mathbf{k}}. \quad (5)$$

For an eigenstate ψ_E of energy E , $\Gamma \psi_E$ is an eigenstate of energy $-E$. We see that for $E = 0$ the Hamiltonian commutes with Γ ; then, since $\Gamma^2 = -\mathbb{1}$, by Kramer's theorem the zero-energy subspace is doubly degenerate. We also note that the operator $\Theta = \Gamma \Pi$, $\Theta^2 = \mathbb{1}$ is an effective time-reversal operator $\Theta H_{\mathbf{k}} \Theta^{-1} = H_{-\mathbf{k}}$.

(3) Reflection symmetries through the plane normal to the x^μ direction, $x^\mu = x, y, z$ are represented by $\Upsilon^\mu = (\sigma_x, \sigma_y, \mathbb{1})$:

$$\Upsilon^\mu H_{\mathbf{k}} \Upsilon^\mu = (-1)^{\rho_\mu} H_{\mathcal{R}^\mu \mathbf{k}}, \quad (6)$$

where the sign $(-1)^{\rho_\mu} = (-1, -1, 1)$ and $\mathcal{R}^\mu \mathbf{k}$ is the reflection of $\mathbf{k}$. Defining $\mathbf{k} = \mathbf{k}_\parallel + \mathbf{k}_\perp$ with parallel and perpendicular components to the plane, $\mathcal{R}^\mu \mathbf{k} = \mathbf{k}_\parallel - \mathbf{k}_\perp$. Thus, our model Hamiltonian is even under a reflection through the xy plane, and odd under reflections through xz and yz planes, since the spin-orbit term breaks the latter symmetries. However, the two broken reflection symmetries result in a spectral symmetry, whereby an eigenstate ψ_E of energy E is mapped to eigenstates $\Upsilon^x \psi_E$ and $\Upsilon^y \psi_E$ of energy $-E$. Note that $\Upsilon^x \Upsilon^y = i\Pi$.

On the surface normal to the Weyl node separation, all reflection symmetries are broken; thus, we only focus on the effective particle-hole symmetry. On surfaces parallel to the Weyl node separation, the reflection symmetry through the surface normal to the Weyl node separation is respected. In our model, this is represented in the spinor space by the identity matrix and a sign reversal of k_z ; thus, as we shall see, our results below respect this symmetry trivially. We shall also see that the effective particle-hole and time-reversal symmetries on these surfaces are spontaneously broken for all self-adjoint extensions of the bulk Hamiltonian.

III. SURFACE BOUND STATES THROUGH SELF-ADJOINT EXTENSION

In order to study the surface states, we consider a semi-infinite geometry terminating at a single two-dimensional plane normal to the x^μ direction. The momentum $\mathbf{k}_\parallel^\mu$ parallel to the plane remains conserved, but the momentum normal to the plane must be treated as the normal component $\hat{\mathbf{p}}_\perp^\mu$ of the momentum operator $\hat{\mathbf{p}} \rightarrow -i\nabla$ in the position basis representation. Correspondingly, the Hamiltonian is the operator $\hat{H}^\mu = H_{\mathbf{k}_\parallel^\mu + \hat{\mathbf{p}}_\perp^\mu}$. The presence of the surface affects the spatial symmetries of the Hamiltonian.

In a bounded or semibounded geometry, the specification of boundary conditions is an integral part of the definition of a self-adjoint operator: different boundary conditions result in different spectral properties. This is because at the physical boundary of the system, different boundary conditions specify different physics at the interface, say with a trivial or topological insulator, a regular or topological metal or semimetal, a superconductor, etc. In this work, we take an agnostic view of what lies on the other side of the interface. Instead, we would like to specify all possible boundary conditions physically admissible. We also pay attention to interfaces that preserve certain bulk symmetries of the system. However, we

note that from a physical point of view, symmetry-breaking boundary conditions are no less interesting, since they can describe interfaces with material that break the corresponding symmetries at the surface.

In a semibounded geometry, in order for the Hamiltonian $\hat{H}^\mu$ to be self-adjoint, its domain $\mathcal{D}(\hat{H}^\mu)$ in the Hilbert space needs to be properly defined. The set of self-adjoint operators is determined fully by von Neumann's deficiency index theorem [29]. This method, while comprehensive, is technical and not physically transparent. Instead, we use a formulation based on a corresponding conserved current derived from the unitary evolution generated by the self-adjoint operator. For two states $\psi, \phi \in \mathcal{D}(\hat{H}^\mu)$, the current defined as

$$j^\mu = i \int_{x_0}^{x^\mu} [\psi^\dagger(\hat{H}^\mu \phi) - (\hat{H}^\mu \psi)^\dagger \phi](\bar{x}^\mu) d\bar{x}^\mu + j_0^\mu, \quad (7)$$

where x_0 is an arbitrary reference point and $j_0^\mu = j^\mu(x_0)$ is local and conserved: it satisfies the continuity equation

$$\frac{\partial \rho}{\partial t} + \frac{\partial j^\mu}{\partial x^\mu} = 0, \quad (8)$$

with $\rho = \psi^* \phi$. By unitarity of time evolution $\frac{d}{dt} \int \rho dx = 0$, where the integral is over the entire system. Since in a semi-infinite geometry, the wave functions vanish at infinity, we must have $j^\mu(\infty) = 0$. Thus,

$$j^\mu(0) = 0. \quad (9)$$

This conserved current is a sesquilinear, Hermitian form of the two arbitrary states ψ and ϕ . In general, Eq. (9) can be expressed as $v_+^\mu[\psi]^\dagger v_+^\mu[\phi] = v_-^\mu[\psi]^\dagger v_-^\mu[\phi]$, where $v_\pm^\mu[\psi]$ is, for our quadratic two-band Hamiltonians, a two-component vector of the wave function and its derivative, normal to the surface, at $x^\mu = 0$. In turn, this yields the boundary condition

$$v_+^\mu[\psi] = U v_-^\mu[\psi], \quad (10)$$

where the unitary matrix $U \in \mathbb{U}(2)$ parametrizes the boundary conditions. We summarize the relevant results in the Appendix.

In the following we parametrize an element of $\mathbb{U}(2)$ as

$$U = e^{im_0} \begin{bmatrix} e^{i\alpha} \cos \delta & -e^{-i\beta} \sin \delta \\ e^{i\beta} \sin \delta & e^{-i\alpha} \cos \delta \end{bmatrix}. \quad (11)$$

As a special case, we note the subset with $\sin \delta = 0$ corresponds to a diagonal $\mathbb{U}(1) \times \mathbb{U}(1)$ subgroup.

The condition to find the spectrum of surface bound states can be obtained by writing the bound state wave function $e^{i\mathbf{k}_\parallel^\mu \cdot \mathbf{x}_\parallel^\mu} \psi_b^\mu(\mathbf{k}_\parallel^\mu, x^\mu)$, with $\mathbf{x}_\parallel^\mu$ the coordinates on the surface, and the bound state

$$\psi_b^\mu(\mathbf{k}_\parallel^\mu, x^\mu) = \frac{1}{\sqrt{\kappa^\mu}} e^{-\kappa^\mu x^\mu} \psi_0^\mu, \quad (12)$$

where κ^μ with $\text{Re} \kappa^\mu > 0$ is the dimensionless inverse confinement length, which may also depend on $\mathbf{k}_\parallel^\mu$. The energy of the bound state is $\pm E_b^\mu$,

$$E_b^\mu(\mathbf{k}_\parallel^\mu, \kappa^\mu) = E_{\kappa^\mu}, \quad (13)$$

where $\kappa^\mu = \mathbf{k}_\parallel^\mu + i\kappa^\mu \hat{\mathbf{x}}^\mu$ and $\hat{\mathbf{x}}^\mu$ is the unit vector in the x^μ direction. The spinor ψ_0^μ has the form

$$\psi_0^\mu \propto \begin{bmatrix} d_{x\kappa^\mu} - i d_{y\kappa^\mu} \\ E_b^\mu - d_{z\kappa^\mu} \end{bmatrix} \quad (14)$$

up to a normalization factor.

In order to determine κ^μ and ψ_0^μ for a given value of energy $E_b^\mu = E$, we invert Eq. (13) and write κ^μ as a function of E . There are two cases that we treat separately:

(1) If there is only one solution with $\text{Re} \kappa^\mu > 0$, the corresponding bound state in Eq. (12) must satisfy the boundary conditions, which we can write as

$$N_+^\mu \psi_0^\mu = U N_-^\mu \psi_0^\mu, \quad (15)$$

where

$$N_\pm^\mu \psi_0^\mu = v_\pm^\mu[\psi_b] \quad (16)$$

depend on κ^μ . The condition to find a nontrivial solution to ψ_0^μ is then

$$\det(N_+^\mu - U N_-^\mu) = 0. \quad (17)$$

(2) If there are more than one possible solutions ψ_{bn}^μ with $\text{Re} \kappa_n^\mu > 0$, indexed by n , we write a general solution as a superposition,

$$\psi_b^\mu = \sum_n c_n \psi_{bn}^\mu, \quad (18)$$

which must satisfy the boundary conditions, which we can write as

$$M_+^\mu c = U M_-^\mu c, \quad (19)$$

where $c = (c_1, c_2, \dots)^\top$, and

$$M_\pm^\mu c = v_\pm^\mu[\psi_b^\mu]. \quad (20)$$

This is then solved for E and c with the condition for solutions to exist given by

$$\det(M_+^\mu - U M_-^\mu) = 0. \quad (21)$$

IV. SURFACE NORMAL TO WEYL NODE SEPARATION

In this case, regardless of our choice of dispersion, the conserved current reads

$$j^z = i(\psi'^\dagger \sigma_z \phi - \psi^\dagger \sigma_z \phi'), \quad (22)$$

where the prime stands for d/dz . Then, for $\psi = (\psi_1, \psi_2)^\top$

$$v_\pm^z[\psi] = \begin{bmatrix} \psi_1 \pm i\psi_1' \\ \psi_2 \mp i\psi_2' \end{bmatrix}, \quad (23)$$

and, denoting the inverse confinement length $\kappa^z = \kappa$, $N_\pm^z = \text{diag}(1 \mp i\kappa, 1 \pm i\kappa)$.

For a diagonal unitary $U \in \mathbb{U}(1) \times \mathbb{U}(1)$, the boundary conditions reduce to two separate sets of conditions on each component of ψ for the self-adjoint extensions of the operator $-d^2/dz^2$ over a semi-infinite geometry. The spectral properties including the surface bound states are previously studied [31]. The full space of $\mathbb{U}(2)$ unitaries is much larger and, therefore, we expect a richer set of bound state spectra.

Before analyzing these spectra, let us characterize the subset of boundary conditions that preserve the relevant symmetries of the bulk. First, under particle-hole operation

$$\Gamma : v_{\pm}^z \mapsto \sigma_y v_{\pm}^{z*}, \quad (24)$$

after which the boundary condition reads $v_{\pm}^z = \sigma_y U^* \sigma_y v_{\pm}^z$. Thus, the boundary condition is symmetric iff $U = \Gamma U \Gamma^{-1}$, which is satisfied for U given in Eq. (11) with $m_0 = 0$ or π .

In this section we shall denote the momentum along the surface $\mathbf{k}_{\parallel}^z = (k_x, k_y, 0) \equiv \mathbf{q}$. We note that upon replacing $\mathbf{k} \rightarrow \kappa = \mathbf{q} + i\kappa\hat{\mathbf{z}}$ in the bulk dispersion, the resulting dispersion E_{κ} does not necessarily represent the surface bound state spectrum. Further conditions and restrictions are imposed by the boundary conditions, from which the existence and the possible values of κ must be extracted. However, assuming such solutions do exist, the functional form of the dispersion is given by E_{κ} .

A. Minimal dispersion

For a given energy of the bound state $E_b^z = E$ with the minimal dispersion, we find $\kappa^2 = \pm\sqrt{E^2 + \lambda^2 q^2} - 1 \in \mathbb{R}$, so there is at most a single solution with $\text{Re}\kappa > 0$, for which $\kappa \in \mathbb{R}$ and, thus, $\kappa^2 = \sqrt{E^2 + \lambda^2 q^2} - 1 > 0$. The bound state energy is then $\pm E_b^z$,

$$E_b^z = \sqrt{(\kappa^2 + 1)^2 + \lambda^2 q^2}. \quad (25)$$

This is a gapped Dirac-like dispersion describing an insulating surface spectrum with minimum gap $\kappa^2 + 1$.

Now, Eq. (17) yields the $\mathbf{q}$ -independent solutions

$$\kappa = \kappa^{(\pm)} = \frac{-\cos \delta \sin \alpha \pm \xi_1 \sqrt{\cos^2 \delta - \cos^2 m_0}}{\cos m_0 + \cos \delta \cos \alpha}, \quad (26)$$

where we have defined a sign $\xi_1 = \text{sgn}(\cos m_0 + \cos \delta \cos \alpha)$. Since only $\kappa \in \mathbb{R}$ is admissible, a necessary condition for the existence of surface states is

$$|\cos \delta| \geq |\cos m_0|. \quad (27)$$

Three separate cases arise, depending on the signs ξ_1 and $\xi_2 = \text{sgn}(\cos \delta \sin \alpha)$. First, if $\xi_1 \xi_2 > 0$, bound states exist only when $|\cos \delta \cos \alpha| > |\cos m_0|$ and $\kappa = \kappa^{(+)}$. Second, if $\xi_1 \xi_2 < 0$ and $|\cos \delta \cos \alpha| > |\cos m_0|$, then bound states exist and $\kappa = \kappa^{(+)}$. However, third, if $\xi_1 \xi_2 < 0$ and $|\cos \delta \cos \alpha| < |\cos m_0|$, then bound states exist for both values of $\kappa = \kappa^{(\pm)}$. In the latter case, bound states form *two* bands of surface energy bands given by Eq. (25)

For particle-hole-symmetric boundary conditions $\sin m_0 = 0$. Then, the necessary condition (27) gives also $\sin \delta = 0$. These boundary conditions form the set of diagonal unitaries $U = \text{diag}(e^{i\alpha_U}, e^{-i\alpha_U})$, where $e^{i\alpha_U} = \text{sgn}(\cos \delta \cos m_0) e^{i\alpha}$. Then, bound states exist only when $\sin \alpha_U < 0$, for which *two degenerate* surface bands exist with $\kappa = -\tan(\alpha_U/2)$.

B. Isotropic dispersion

For the isotropic dispersion, denoting $E_b^z = E$ and $\kappa^z = \kappa$, we find two possible solutions $\kappa_{\pm}^2 = q^2 - 1 \pm \sqrt{E^2 + \lambda^2 q^2}$. When κ_-^2 becomes negative and therefore unphysical, there is only one possible value of κ with $\text{Re}\kappa > 0$. This value is

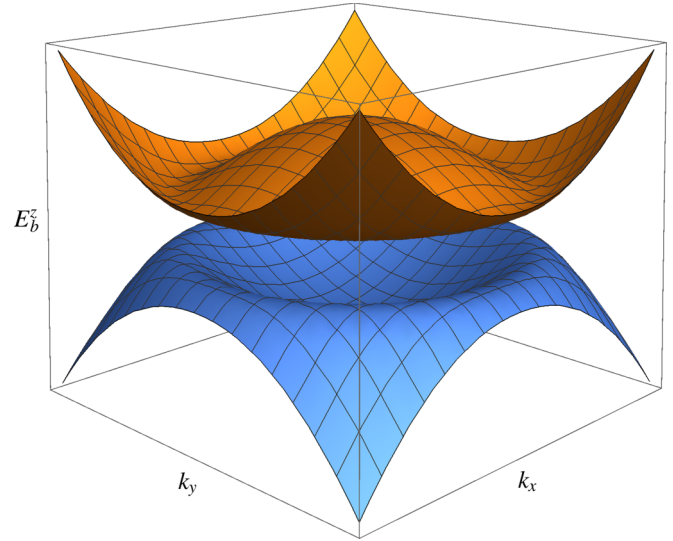


FIG. 1. The bound state dispersion E_b^z on the surface normal to the Weyl node separation with momentum (k_x, k_y) along the surface, in units of half the separation of the Weyl nodes, and dimensionless spin-orbit coupling $\lambda^2 < 2(\kappa^2 + 1)$, where κ is the inverse confinement length of surface bound states.

found from the boundary conditions leading to Eq. (26) and the subsequent analysis. A sufficient condition for $\kappa_-^2 < 0$ is

$$q < q_* = \lambda/2 + \sqrt{(\lambda/2)^2 + 1}. \quad (28)$$

Then, the bound state energy is $\pm E_b^z$,

$$E_b^z = \sqrt{(q^2 - \kappa^2 - 1)^2 + \lambda^2 q^2}. \quad (29)$$

This spectrum has a direct insulating gap, obtained at $q = 0$ when $\lambda^2 > 2(\kappa^2 + 1)$ and at $q_m = \sqrt{\kappa^2 + 1 - \lambda^2/2}$ when $\lambda^2 < 2(\kappa^2 + 1)$. As shown in Fig. 1, in the latter case the surface spectrum has a Mexican-hat shape with a minimum energy $E_b^z = \lambda\sqrt{q_m^2 + (\lambda/2)^2}$ at a circle with radius q_m and a local maximum at $q = 0$, $E_b^z = \kappa^2 + 1$. Thus for chemical potentials between these local minimum and maximum, the Fermi surface consists of an electronlike and a holelike pockets.

When both $\kappa_{\pm}^2 > 0$, we find two possible solutions with $\text{Re}\kappa_{\pm} > 0$. In this case, taking $q_x + iq_y \equiv qe^{i\phi}$, we first find the spinor ψ_0^z from Eq. (14) to be

$$\psi_{0\pm}^z \propto \begin{bmatrix} \lambda q e^{-i\phi} \\ E + \kappa_{\pm}^2 + 1 \end{bmatrix}. \quad (30)$$

Then, taking a general superposition in Eq. (18), we find

$$M_{\pm} = \begin{bmatrix} \lambda q e^{-i\phi}(1 \mp i\kappa_{+}) & \lambda q e^{-i\phi}(1 \mp i\kappa_{-}) \\ (E + \kappa_{+}^2 - 1)(1 \pm i\kappa_{+}) & (E + \kappa_{-}^2 - 1)(1 \pm i\kappa_{-}) \end{bmatrix}. \quad (31)$$

Finally, E is found as a solution to Eq. (21).

For general U , the above procedure is tedious and not very instructional. Indeed, there may not even be any solutions depending on the choice of U , i.e., the form of the boundary conditions. To illustrate this point, let us take a simple example $U = -\mathbb{1}$ corresponding to the boundary condition

$\psi(z=0) = 0$. Then,

$$M_+ - UM_- = M_+ + M_- \quad (32)$$

$$= -2i \begin{bmatrix} \lambda q e^{-i\phi} & \lambda q e^{-i\phi} \\ E + \kappa_+^2 - 1 & E + \kappa_-^2 - 1 \end{bmatrix}. \quad (33)$$

Thus, $\det(M_+ - UM_-) = 0$ only when $\kappa_+ = \kappa_-$, i.e., $q = 0$, which is outside the acceptable range of q . A similar result follows for $U = \mathbb{1}$, which corresponds to the boundary condition $\psi'(z=0) = 0$.

This is a remarkable and unconventional result: for boundary conditions as commonly used as the vanishing of the wave function or its derivative on the plane, we see that not only the spectrum has an uncommon Mexican-hat shape for a range of spin-orbit couplings, but that no surface bound states exist in a “forbidden” range of momenta in the surface Brillouin zone.

V. SURFACE PARALLEL TO THE WEYL NODE SEPARATION: FERMI ARCS

We take the surface normal to the y direction below and denote the momentum along the surface $\mathbf{k}_\parallel^y = (k_x, 0, k_z) \equiv \mathbf{p}$.

A. Minimal dispersion

With the minimal dispersion, the conserved current takes the form

$$j^y = \lambda \psi^\dagger \sigma_y \psi. \quad (34)$$

This current yields, in fact, $v_\pm[\psi] = \psi_1 \pm i\psi_2$, which are complex numbers instead of spinors. Thus, the boundary conditions in this case are characterized by a single $\mathbb{U}(1)$ phase [29,32]

$$\psi_1 + i\psi_2 = e^{i\gamma}(\psi_1 - i\psi_2). \quad (35)$$

Under particle-hole operation, $\Gamma : v_\pm \mapsto \mp v_\mp^*$ and $e^{i\gamma} \mapsto -e^{i\gamma}$. Thus, all permissible boundary conditions supporting a self-adjoint extension of the Hamiltonian with the minimal dispersion break particle-hole symmetry on the surface parallel to Weyl node separation [33].

Denoting the inverse confinement length $\kappa^y = \kappa$ and the bound state energy $E_b^y = E$, we have $E^2 = (k_z^2 - 1)^2 + \lambda^2(k_x^2 - \kappa^2)$. So again, for a given energy, there is at most one solution with $\text{Re}\kappa > 0$. Solving for $\psi_0^y \propto \begin{bmatrix} 1 \\ -\rho \end{bmatrix}$ in the boundary condition equation, we have

$$\frac{1 - i\rho}{1 + i\rho} = e^{i\gamma} \Rightarrow \rho = \tan \frac{\gamma}{2}. \quad (36)$$

Note that ρ is independent of $\mathbf{p}$ and depends only on the choice of boundary condition. We can then find κ from Eq. (14), which yields

$$\rho = \frac{\lambda\zeta - E}{\lambda(k_x + \kappa)}, \quad (37)$$

with $\lambda\zeta \equiv k_z^2 - 1$.

The surface state spectrum is then found by solving Eq. (37). The spectrum can be rewritten as $(\lambda\kappa)^2 + E^2 = (\lambda k_x)^2 + (\lambda\zeta)^2 \equiv s^2$; defining two angular variables $-\pi < \theta < \pi$ and $-\pi/2 < \varphi < \pi/2$ as $\lambda k_x = s \cos \theta$, $\lambda\zeta = s \sin \theta$ and $\lambda\kappa = s \cos \varphi$, $E = s \sin \varphi$, we have $\tan(\gamma/2) = \rho = \tan[(\theta - \varphi)/2]$.

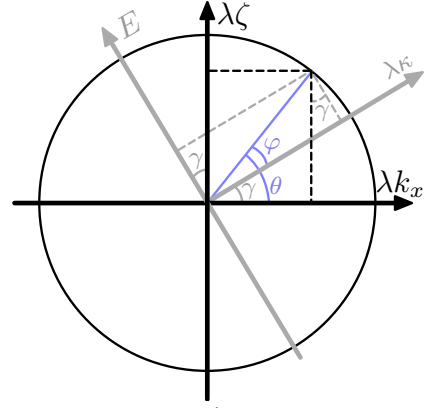


FIG. 2. The geometric structure of the surface state spectra normal to the y direction, parallel to the direction of Weyl node separation in the z direction. A pair of vectors $(\lambda\kappa, E)$ and $(\lambda k_x, \lambda\zeta)$ are related by having the same length. For the minimal dispersion, $\lambda\zeta = k_z^2 - 1$ and the angle γ between these two vectors is fixed by the boundary condition. For the isotropic dispersion, there is in general two possible solutions for the inverse confinement length κ ; in this case, the algebraic relationship between the boundary condition and the solutions is discussed in the text.

Thus, $\theta - \varphi = \gamma$ is fixed by the boundary condition. We then find for $E = E_b^y$,

$$\lambda k_x = \cot \gamma (k_z^2 - 1) - \csc \gamma E. \quad (38)$$

with the condition

$$\lambda\kappa = \tan \gamma E + \sec \gamma \lambda k_x \quad (39)$$

$$= \sin \gamma (k_z^2 - 1) + \cos \gamma \lambda k_x > 0. \quad (40)$$

The geometry of angular variables is shown in Fig. 2. This is the Fermi arc on the surface parallel to the bulk Weyl nodes. At $E = 0$, it connects the Weyl nodes projected on this surface at $k_x = 0, k_z = \pm 1$. At $E = 0$, $\lambda\kappa = (k_z^2 - 1)/\sin \gamma > 0$, so the Fermi arc is restricted to $(|k_z| - 1)\xi_\gamma > 0$, where $\xi_\gamma = \text{sgn} \sin \gamma$: for $\xi_\gamma < 0$, it stays within $|k_z| < 1$; for $\xi_\gamma > 0$, it extends to the Brillouin zone edges. At finite E , these ranges vary, dividing the Brillouin zone into allowed and “forbidden” regions where surface bound states are absent. These spectra are depicted in Fig. 3.

Note that, in contrast to the surface normal to the Weyl node separation, the boundary conditions on the parallel surface involve both E and κ . Together with the eigenvalue equation, the solutions then only allow for a single band. Thus, the bound state dispersion on this surface has a single slope, i.e., it is “chiral.” This is consistent with the fact that particle-hole symmetry is broken for all boundary conditions of the self-adjoint extensions of the bulk Hamiltonian. (We note that, as anticipated, reflection symmetry in the z direction is trivially satisfied since the dispersion is even in k_z .) This chiral dispersion reflects the topological nature of these surface states.

B. Isotropic dispersion

Now, the conserved current reads

$$j^y = i(\psi^\dagger \sigma_z \psi - \psi^\dagger \sigma_z \psi') + \lambda \psi^\dagger \sigma_y \psi, \quad (41)$$

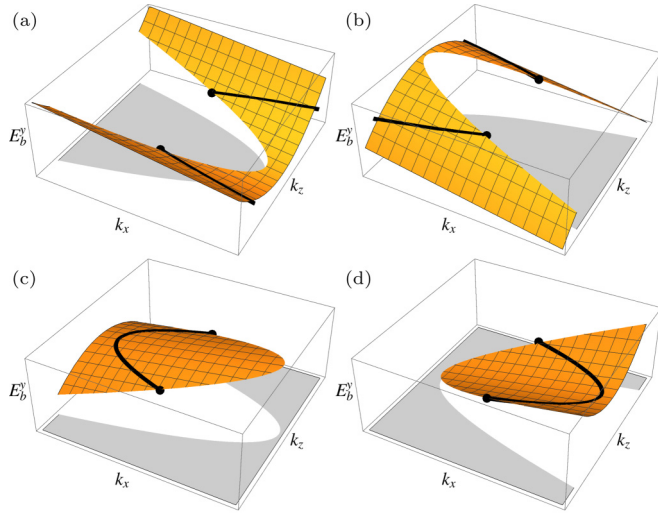


FIG. 3. The spectrum on the surface parallel to the Weyl node separation. The shaded region shows the forbidden region where the confinement length diverges and surface bound states are absent. Four different types of boundary conditions are characterized by the angle γ , see Eq. (36), (a) $0 < \gamma < \pi/2$, (b) $\pi/2 < \gamma < \pi$, (c) $\pi < \gamma < 3\pi/2$, and (d) $3\pi/2 < \gamma < 2\pi$. The black circles show the projection of the Weyl nodes on the surface and the solid black curve connecting them is the Fermi arc at $E = 0$.

which yields

$$v_{\pm}^y[\psi] = \begin{bmatrix} \psi_1 \mp i(\psi_2 + 2\psi_1'/\lambda) \\ \psi_2 \pm i(\psi_1 + 2\psi_2'/\lambda) \end{bmatrix}. \quad (42)$$

Writing the bound states $\psi_b^y \propto [_{-\rho}^1]e^{-\kappa y}$ with energy $E_b^y = E$, we find two possible solutions

$$\kappa_{\pm}^2 = p^2 - 1 + \frac{1}{2}\lambda^2 \pm \sqrt{E^2 + \lambda^2[k_z^2 - 1 + (\lambda/2)^2]}, \quad (43)$$

and $\rho_{\pm} = \frac{p^2 - 1 - \kappa_{\pm}^2 - E}{\lambda(k_x + \kappa_{\pm})}$, where $\mathbf{p} = (k_x, 0, k_z)$. Note that if we define $\zeta_{\pm} = -\lambda/2 \mp \sqrt{(E/\lambda)^2 + k_z^2 - 1 + (\lambda/2)^2}$, we have $(\lambda\kappa_{\pm})^2 + E^2 = (\lambda k_x)^2 + (\lambda\zeta_{\pm})^2 \equiv s_{\pm}^2$ and $\rho_{\pm} = \frac{\lambda\zeta_{\pm} - E}{\lambda k_x + \lambda\kappa_{\pm}}$. Then, for each solution the pair of vectors $(\lambda k_x, \lambda\zeta_{\pm})$ and $(\lambda\kappa_{\pm}, E)$, with (complex) lengths $s_{\pm}$ and (complex) angles $\theta_{\pm}$ and $\varphi_{\pm}$ constrained to $s_+ \sin \varphi_+ = s_- \sin \varphi_- = E$ and $s_+ \cos \theta_+ = s_- \cos \theta_- = \lambda k_x$, admit a similar geometric structure as shown in Fig. 2.

Now, since $E^2 + \lambda^2[k_z^2 - 1 + (\lambda/2)^2] = E^2 + \lambda^2 k_z^2 - \frac{1}{4}\lambda^2(4 - \lambda^2) \equiv W$ is not always positive, even for small p , we may have two solutions with $\text{Re}\kappa_{\pm} > 0$. Indeed, for small p and E there is a finite range of λ where this is true: for example, taking $E = 0$ for a given k_z , $W < 0$ for $(\lambda/2)^2 < 1 - k_z^2$ and we find two permissible solutions for the bound states. In this case,

$$M_{\pm} = \begin{bmatrix} 1 \pm i(\rho_+ + \frac{2}{\lambda}\kappa_+) & 1 \pm i(\rho_- + \frac{2}{\lambda}\kappa_-) \\ -\rho_+ \pm i(1 + \frac{2}{\lambda}\kappa_+\rho_+) & -\rho_- \pm i(1 + \frac{2}{\lambda}\kappa_-\rho_-) \end{bmatrix}. \quad (44)$$

In general, the spectrum can now be found from Eq. (21). However, this is tedious and not very instructional. In the following we address some special cases.

First, we show that for certain boundary conditions, the Fermi arcs are straight lines connecting the Weyl nodes. In particular, assuming $|k_z| \leq 1$ and taking $E = k_x = 0$, we have

$$\kappa_{\pm} = \lambda/2 \pm \sqrt{k_z^2 - 1 + (\lambda/2)^2} = -\zeta_{\pm} \quad (45)$$

and $\rho_{\pm} = -1$. Then, $M_{\pm} = [\frac{1 \pm i\omega}{1 \mp i\omega} \quad \frac{1 \mp i\omega}{1 \pm i\omega}]$, where $\omega = \frac{2}{\lambda}\sqrt{k_z^2 - 1 + (\lambda/2)^2}$. Now, taking U as in Eq. (11) and solving Eq. (21), we find

$$\sin \delta \sin \beta = \sin m_0. \quad (46)$$

For example, as opposed to the minimal dispersion, $U = -1$, corresponding to $\psi(0) = 0$, and $U = 1$, corresponding to $\psi'(0) + (\lambda/2)\sigma_x \psi(0) = 0$, support straight Fermi arcs connecting the two Weyl nodes.

Next, we shall determine the spectrum for a specific choice of boundary conditions given by $U = -1$, i.e., $\psi(0) = 0$; then $M_+ - U M_- = M_+ + M_- = [\frac{2}{-\rho_+} \quad \frac{2}{-\rho_-}]$ and Eq. (21) yields $\rho_+ = \rho_- \equiv \rho$ and $c_+ = -c_-$. Using the algebraic relationship defining $\rho_{\pm}$ and $\kappa_{\pm}$ we find

$$\lambda\rho = -(\kappa_+ + \kappa_-), \quad (47)$$

$$E = \frac{\lambda^2}{2}(\rho^2 - 1) - \rho\lambda k_x. \quad (48)$$

Together with Eq. (43), these equations determine the spectrum. As must be the case, we see that for $E = k_x = 0$ $\rho = -1$ and the equations are satisfied for any $|k_z| < 1$ (this is necessary for having two solutions as discussed above). For small E and k_x , we may expand $\kappa_{\pm}$ to find $\rho = -1 + [E^2 - (\lambda k_x)^2]/[2\lambda^2(1 - k_z^2)] + O(k_x^2 E^2)$. Thus, up to this order, the surface state spectrum $E = E_b^y$ is given by

$$E_b^y = \lambda k_x. \quad (49)$$

Indeed, we can show that this spectrum is correct to all orders. Taking $E = \lambda k_x$ and working backwards the two solutions $\kappa_{\pm} = \lambda/2 \pm i\sqrt{1 - (\lambda/2)^2 - p^2}$, where we have assumed $p^2 < 1 - (\lambda/2)^2$ in order to have $\text{Re}\kappa_{\pm} > 0$. Thus, $\rho = -1$, and we see that Eq. (48) is satisfied automatically. The wave function is found as

$$\psi_b^y(\mathbf{p}, y) = \sqrt{\lambda[1 + (\lambda/2\chi)^2]}e^{-\frac{\lambda}{2}y} \sin(\chi y) \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad (50)$$

with $\chi = \sqrt{1 - (\lambda/2)^2 - p^2}$. Note that the surface states all have the same confinement length $2/\lambda$ and a standing-wave modulating factor with the wave number χ that depends on the momentum $\mathbf{p}$ on the surface. We note that particle-hole symmetry is broken for these surface states since the dispersion has a preferred chirality.

The mapping of the general boundary conditions under particle-hole operation for the isotropic dispersion is more complicated than before: we have

$$\Gamma : v_{\pm} \mapsto S_{\pm}^+ v_{\pm}^* + S_{\pm}^- v_{\pm}^*, \quad (51)$$

with $S_{\pm}^{\xi} = \frac{1}{2}(\mp 1 - \xi\sigma_x + \sigma_y + i\sigma_z)$. Thus, a particle-hole symmetric boundary condition must satisfy

$$U S_{\pm}^+ U^* + U S_{\pm}^- - S_{\pm}^+ U^* = S_{\pm}^-. \quad (52)$$

As expected from the chiral surface states we found above, it is easy to see that $U = -\mathbb{1}$ does not satisfy this condition: $S_-^+ - (S_-^- - S_+^+) = \frac{1}{2}(\mathbb{1} - \sigma_x + \sigma_y + i\sigma_z) - (\mathbb{1} + \sigma_x) = \frac{1}{2}(-\mathbb{1} - 3\sigma_x + \sigma_y + i\sigma_z) \neq S_+^-$. Similarly, for $U = \mathbb{1}$: $S_-^+ + (S_-^- - S_+^+) = \frac{1}{2}(\mathbb{1} - \sigma_x + \sigma_y + i\sigma_z) + (\mathbb{1} + \sigma_x) = \frac{1}{2}(3\mathbb{1} + \sigma_x + \sigma_y + i\sigma_z) \neq S_+^-$. Our numerical search for solutions to this set of unitaries returned a null result; thus, we conclude that, as for the minimal dispersion, all self-adjoint boundary conditions of the isotropic bulk Hamiltonian also break particle-hole symmetry on the surface parallel to the Weyl node separation.

We close this section by considering the relationship between the surface bound states of the minimal and the isotropic bulk dispersions. One may expect the two cases to be related at large values of λ where the quadratic terms in k_x and k_y become negligible compared to the linear spin-orbit interaction. However, this connection should be obtained at the level of boundary conditions, not the final solutions.

For example, note that the solutions in Eq. (36) with the minimal dispersion and, say, Eq. (50) with the isotropic dispersion have different spinor structures independent of λ . This can be seen also by the fact that the Fermi arcs of the minimal dispersion are generally curved while the Fermi arcs of the solution Eq. (50) are straight lines. This is because the boundary conditions leading to Eqs. (36) and (50) are not simply related. Indeed, the condition $\psi(0) = 0$ leading to Eq. (50) is not allowed for the minimal dispersion: it is not satisfied for any value of γ in Eq. (35).

At the level of boundary conditions, we may note that for large λ the boundary conditions obtained from Eq. (42) reduce to Eq. (35) for $\delta = 0, \alpha = \pm\pi/2$ and $m_0 = -\gamma - \alpha$. A straight Fermi arc is obtained for the minimal dispersion only when $\gamma = \pm\pi/2$, which corresponds to either $\delta = m_0 = 0, \alpha = -\gamma$ or $\delta = 0, m_0 = \mp\pi, \alpha = \gamma$. In any case, according to Eq. (46), the isotropic dispersion for these boundary conditions also supports a straight Fermi arc.

VI. CONCLUSION

We have studied the most general form of boundary conditions for a simple Hamiltonian of a Weyl semimetal with a single pair of Weyl nodes in the continuum limit. Our approach, based on self-adjoint extensions of the bulk Hamiltonian in bounded geometries, allows us to determine all physically possible boundary conditions and their respective surface spectra. These boundary conditions describe different physical interfaces at the boundary of the Weyl semimetal with other phases of matter. We also studied the role of bulk symmetries. In particular, we find that on the surface parallel to the direction of Weyl node separation, all physical boundary conditions break particle-hole symmetry and lead to chiral surface states.

Using a minimal as well as an isotropic bulk dispersion, we determined physical surface spectra and found various types of spectra, including Fermi arcs, depending on the boundary conditions. On the surface normal to the direction of Weyl node separation, interesting gapped spectra arise for a wide family of boundary conditions. For example, a dispersion in the shape of a Mexican hat exists on this surface in a restricted region of the surface Brillouin zone. In the presence of interactions, or by engineering the local potential to induce coexisting particle

and hole Fermi surfaces, such a spectrum could support novel correlated phases of matter at the surface. On the surface parallel to the Weyl node separation, we determined the general conditions giving the surface bound states and Fermi arcs. We found chiral surface bound states in restricted parts of the surface Brillouin zone; the chiral nature of the surface states is consistent with their topological nature.

Some interesting directions for future work are finding concrete models that realize given boundary conditions and studying the effects of interactions or potential profiles in cases of unusual surface state spectra. Another interesting direction is to study the general self-adjoint boundary conditions in other topological material, such as type-II Weyl semimetals [34,35].

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APPENDIX: SELF-ADJOINT EXTENSIONS

The distinction between self-adjoint and Hermitian operators is a subtlety that is usually ignored in introductions to quantum mechanics [31]. One reason for this is that the two definitions coincide for nonsingular Hamiltonians acting on systems that extend to infinity in all directions. When either of these conditions fail, the two are not the same, and a further analysis is needed. There is a strong physical basis for requiring that observables should correspond with self-adjoint operators. One key result comes from Stone's theorem [36], which states that a transformation of the form $\exp(itA)$ is unitary for real t if and only if A is self-adjoint.

The difference between self-adjoint and Hermitian operators is a constraint on the functions that they can be defined on. All self-adjoint operators are Hermitian, but the converse is not true. Unbounded operators (derivatives in our case) have some properties with no analog in finite-dimensional linear algebra. One such result is the Hellinger-Toeplitz theorem [36], which states that unbounded, Hermitian operators cannot be defined on the entire Hilbert space $\mathcal{H}$.

With this context in mind, the definitions will be more transparent. Recall that an operator A is Hermitian if

$$\langle \psi | A \phi \rangle = \langle A \psi | \phi \rangle \quad (\text{A1})$$

for any elements ψ and ϕ in the domain $\mathcal{D}(A)$ of A . The adjoint $A^\dagger$ of A is defined through

$$\langle \psi | A \phi \rangle = \langle A^\dagger \psi | \phi \rangle \quad (\text{A2})$$

for all $\phi \in \mathcal{D}(A)$ and ψ in the adjoint domain $\mathcal{D}(A^\dagger)$. A self-adjoint operator A is a Hermitian operator with $\mathcal{D}(A) = \mathcal{D}(A^\dagger)$. In the general case, the domains of A and $A^\dagger$ are unrelated in the sense that one does not necessarily contain the other. However, when A is Hermitian, the domain is always contained in the adjoint domain, that is $\mathcal{D}(A) \subseteq \mathcal{D}(A^\dagger)$. We can interpret self-adjoint operators as maximally defined Hermitian operators.

This suggests that there is a method of extending the domain of a Hermitian operator to promote it to a self-adjoint operator. von Neumann's theorem allows us to determine when this is true and classify all possible extensions.

1. Deficiency indices and von Neumann's theorem

The procedure outlined by von Neumann's theorem has the benefit that it may be applied to any Hermitian operator A . In the first step, we compute its adjoint $A^\dagger$ and its domain $\mathcal{D}(A^\dagger)$. The second step is to pick two values $+i\eta_+$ and $-i\eta_-$ laying in the upper and lower half of the complex plane, respectively. With this choice, we solve the eigenvalue equations

$$A^\dagger\psi = \pm i\eta_\pm\psi \quad (\text{A3})$$

and determine the eigenspaces $\mathcal{E}_\pm \subseteq \mathcal{D}(A^\dagger)$ corresponding to $\eta_\pm$. An important result states that the deficiency indices $n_\pm = \dim \mathcal{E}_\pm$ do not depend on the choice of $\eta_\pm$.

Now, von Neumann's deficiency index theorem states [36] that iff $n_+ = n_- = n$, then A has (infinitely many) self-adjoint extensions A_U , parametrized by a $n \times n$ unitary map $U : \mathcal{E}_+ \rightarrow \mathcal{E}_-$, and defined over the domain $\mathcal{D}(A_U) = \{\phi_U = \phi + \psi_+ + U\psi_+\}$, where $\phi \in \mathcal{D}(A)$ and $\psi_+ \in \mathcal{E}_+ \subset \mathcal{D}(A^\dagger)$, with the action $A_U\phi_U = A\phi + i\eta_+\psi_+ - i\eta_-U\psi_+$. In a bounded geometry, each self-adjoint extension corresponds to a boundary condition parametrized by U .

2. Conserved-current formulation

While von Neumann's deficiency index method is general, it is not physically transparent. A more intuitive picture of self-adjoint extensions is given by a conserved current defined, up to a constant, by [29]

$$j_A = i \int^x [\psi^\dagger(A\phi) - (A\psi)^\dagger\phi](\bar{x}) d\bar{x}, \quad (\text{A4})$$

as in Eq. (9). In general, this current can be written as a sesquilinear form of ψ and ϕ as $j = v[\psi]^\dagger J_A v[\phi]$, where $v[\psi]$ is a vector of the wave function and its derivatives and J_A is a finite Hermitian matrix. In a semi-infinite geometry, the conservation condition $j_A(0) = 0$ can then be written algebraically as

$$v_+[\psi(0)]^\dagger v_+[\phi(0)] = v_-[\psi(0)]^\dagger v_-[\phi(0)], \quad (\text{A5})$$

where $v_+[\psi] \oplus v_-[\psi] := (d_+ \oplus d_-)T v[\psi]$ and T is a unitary matrix that diagonalizes J_A as $T J_A T^\dagger = (d_+ \oplus d_-)^2$, with $d_\pm = \pm d_\pm^\dagger$ diagonal. In this diagonal basis, d_+ (d_-) are the square roots of J_A projected to the subspace of positive (negative) eigenvalues of J_A . This condition states that the inner product of $v_\pm$ must be preserved regardless of the states ψ and ϕ . Thus, the existence of self-adjoint extensions of A is equivalent to the condition

$$v_+[\psi(0)] = U v_-[\psi(0)], \quad (\text{A6})$$

for a unitary matrix U [29].

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On the role of self-adjointness in the continuum formulation of topological quantum phases

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Abstract

Topological quantum phases of matter are characterized by an intimate relationship between the Hamiltonian dynamics away from the edges and the appearance of bound states localized at the edges of the system. Elucidating this correspondence in the continuum formulation of topological phases, even in the simplest case of a one-dimensional system, touches upon fundamental concepts and methods in quantum mechanics that are not commonly discussed in textbooks, in particular the self-adjoint extensions of a Hermitian operator. We show how such topological bound states can be derived in a prototypical one-dimensional system. Along the way, we provide a pedagogical exposition of the self-adjoint extension method as well as the role of symmetries in correctly formulating the continuum, field-theory description of topological matter with boundaries. Moreover, we show that self-adjoint extensions can be characterized generally in terms of a conserved local current associated with the self-adjoint operator.

I. INTRODUCTION

Phases of matter are operationally distinguished by their relevant physical observables. In quantum mechanics, these observables correspond to linear, self-adjoint, and gauge-invariant operators acting on a properly specified Hilbert space, whose elements represent the physical states of the system.¹ In this formalism, measurable values of an observable are identified with the eigenvalues of the corresponding operator, and since the result of a measurement is associated with a real number it is sensible to impose the condition of self-adjointness to guarantee real-valued eigenvalues. When the Hilbert state space is infinite-dimensional, some operators may be unbounded and, therefore, need to be treated with special care; in particular, determining the proper set of boundary conditions that establishes their self-adjointness becomes a crucial step in their definition.² Specifically, for an unbounded operator to be self-adjoint, it is necessary but not sufficient for it to be Hermitian (symmetric). Therefore, a naive approach to analyzing the eigenvalues and eigenvectors of Hermitian operators without properly specifying the self-adjoint boundary conditions can lead to unexpected paradoxes,² resulting in wrong physical conclusions.

These considerations become quite important for the study of topological phases of matter,^{3,4} where the properties of the physical system near boundaries are closely linked with those of the bulk. In view of the increased interest in topological phases, it is a goal of this work to illustrate the correct implementation of the theory of self-adjoint extensions of operators to the continuum, or field-theory, description of topological quantum matter. Fortunately, there is already a well-developed theory of self-adjoint extensions of Hermitian operators;^{5,6} however, its application to, and consequences for, the continuum description of topological phases remain obscure. We believe this is in part due to the mathematically abstract nature of this theory. Thus, we also aim to provide a formulation of the theory of self-adjoint extensions that is more natural for physicists.

We will illustrate this method in the context of a concrete and simple model of a topological phase. From a pedagogical standpoint, topological phases provide a very natural setting for learning and using self-adjoint extensions. It is a topic that brings together several strands of modern physics and mathematics: many-body quantum systems; symmetries of a physical system; topology; and functional analysis. We present an accessible exposition of these subjects in the context of a simple model of a ferromagnetic insulator with spin-orbit

interactions. We also clarify the role of symmetry and boundary conditions in self-adjoint extensions of an operator. In particular, we show that the notion of bulk-boundary correspondence, i.e., a relationship between the topological properties of the bulk phase and the character of its boundary states (see, e.g., Chap. 6 of Ref. 3), depends on the choice of self-adjoint extension. Finally, we provide a new and intuitive physical interpretation of self-adjoint extensions in terms of a conserved current associated with the self-adjoint operator.

The paper is organized as follows. In Sec. II we briefly explain the notions of a topological state of matter and its concomitant boundary states. To illustrate these concepts, in Sec. III we introduce a model of a one-dimensional lattice that exhibits normal and topological phases over a range of parameters. In Sec. IV we present a pedagogical derivation of the continuum, field-theory description of this model with boundaries, and point out the difficulty of dealing with unbounded operators. In Sec. V we review von Neumann's theory of self-adjoint extensions of a Hermitian operator. In Sec. VI we use this theory to obtain the self-adjoint extensions of the continuum model Hamiltonian of Sec. IV and its topological bound states. We also discuss the effect of a symmetry operation on the self-adjoint extension. In Sec. VII we recast the self-adjoint extensions of arbitrary Hermitian operators in terms of a conserved, spatially local current. We conclude in Sec. VIII. Technical details are presented in four appendices.

II. TOPOLOGICAL QUANTUM PHASES AND BOUNDARY STATES

The study of topological phases of quantum systems has flourished in recent years, thanks to theoretical and experimental discoveries of several families of such phases in artificial nanostructures as well as bulk crystals.^{3,4} In contrast to the phases of matter characterized by spontaneously broken symmetries, such as ferromagnets vs. paramagnets or solids vs. liquids,⁷ topological phases of electronic matter are characterized by the nontrivial topology of their electrons' wave functions. In particular, a topological phase cannot be distinguished from a normal phase by probing its bulk electronic properties locally. For example, a topological insulator does not allow the passage of electric current through it just like a normal insulator; however, as one changes the momentum of the electrons, in a topological insulator, their wave functions exhibit a nontrivial twist. This twist cannot be undone by changing the

parameters of the system unless the system is brought to a critical phase at which it becomes a metal.⁸ This characterization also implies that at the boundary between a topological phase and a normal phase, the electronic motion must undergo a drastic change. For instance, the electronic states at the boundary between a topological and a normal insulator are metallic. In other words, one expects that at the boundaries of the system in a topological phase there are bound states, protected by symmetry, that emerge because of the nontrivial topology of the bulk. This statement is commonly referred to as the bulk-boundary correspondence.

Though crystals are described microscopically in terms of their lattice structure, it is often more convenient to describe these systems in a continuum, field-theory, representation. Such a description is relevant for the long-distance behavior of the system, which is what determines its quantum phases. It is also a more general description since different lattice systems can end up having the same continuum description and, thus, the same long-distance physics. As is generically the case in quantum systems, continuum descriptions involve unbounded operators such as the linear momentum. Usually this does not present a major obstacle since one can regularize such operators by restricting their action to square-integrable wave functions that vanish sufficiently fast at infinity. However, in order to illustrate the bulk-boundary correspondence in a topological phase, one must study the properties of the system near the boundaries. In this case, one needs to regularize the unbounded operators differently. Intuitively, this regularization can be seen to involve the boundary conditions of the electronic wave functions. Mathematically, it requires the notion of self-adjoint extensions of unbounded Hermitian operators.^{2,9,10}

In order to illustrate these concepts, in the next section we introduce a simple lattice model in one spatial dimension, which exhibits a normal and a topological insulating phase. We derive the continuum Hamiltonian description of this model and study the spectral properties of the corresponding field theory. We shall see that correctly identifying the phases of the system in this continuum formulation requires the proper identification of boundary conditions related to the self-adjointness of the Hamiltonian operator.

III. A TOPOLOGICAL QUANTUM WIRE

Consider a system of spin-1/2 fermions moving along a line with a periodic array of potential wells, for example electrons moving through a chain of N ions, subject to a magnetic

field and an internal spin-orbit interaction. When the separation between the ions a is large enough so that only the quantum tunneling between nearest ions is appreciable, one can model this system as a discrete chain with the Hamiltonian,⁴

$$\hat{H} = \sum_{r=1}^N \left[\frac{\mu}{2} \left(\hat{c}_{\uparrow r}^\dagger \hat{c}_{\uparrow r} - \hat{c}_{\downarrow r}^\dagger \hat{c}_{\downarrow r} \right) + w \left(\hat{c}_{\uparrow r+1}^\dagger \hat{c}_{\uparrow r} - \hat{c}_{\downarrow r+1}^\dagger \hat{c}_{\downarrow r} \right) + \frac{\lambda}{2} \left(\hat{c}_{\downarrow r+1}^\dagger \hat{c}_{\uparrow r} - \hat{c}_{\uparrow r+1}^\dagger \hat{c}_{\downarrow r} \right) \right] + \text{h.c.} \quad (1)$$

Here, “+h.c.” means one must add the Hermitian conjugate of all the previous terms. We are using a second-quantized notation of the creation and annihilation operators in the Fock space, $\hat{c}_{sr}^\dagger$ and $\hat{c}_{sr}$, whose action is to create and remove, respectively, an electron at ion site $r \in \{1, 2, \dots, N\}$ with spin $s \in \{\uparrow, \downarrow\}$ along the direction of the magnetic field. These operators obey the anti-commutation relations $\hat{c}_{sr} \hat{c}_{s'r'} + \hat{c}_{s'r'} \hat{c}_{sr} = 0$ and $\hat{c}_{sr} \hat{c}_{s'r'}^\dagger + \hat{c}_{s'r'}^\dagger \hat{c}_{sr} = \delta_{ss'} \delta_{rr'}$, where δ is the Kronecker delta. The operator product $\hat{c}_{sr}^\dagger \hat{c}_{sr}$ counts how many electrons (0 or 1) of spin s are at site r . The operator product $\hat{c}_{s'r+1}^\dagger \hat{c}_{sr}$ displaces an electron of spin s from site r to one of spin s' at site $r+1$. In this way, we can see that the parameter μ in the model reflects the Zeeman energy splitting in the magnetic field; w is an energy scale related to the strength of the spin-dependent quantum tunneling between two nearest ions; and λ is related to the strength of the spin-orbit interaction. (Note the absence of a spin-independent tunneling term, and the fact that the sum in Eq. (1) does not include terms with $\hat{c}_{sN+1}$ and $\hat{c}_{sN+1}^\dagger$.) When the number of electrons is N , the system may represent a ferromagnetic insulator with spin-orbit coupling. For simplicity, we shall take all the parameters to be real and w and λ to be positive numbers.

Note that this model has the following property: upon flipping the spin $\hat{c}_{\uparrow r} \mapsto \hat{c}_{\downarrow r}$ and $\hat{c}_{\downarrow r} \mapsto \hat{c}_{\uparrow r}$, one finds $\hat{H} \mapsto -\hat{H}$. Denoting the operation with the unitary map $\hat{S} = (\hat{S}^\dagger)^{-1}$, we can express this as $\hat{S} \hat{H} \hat{S}^\dagger = -\hat{H}$. This means that if we flip the spins in any energy eigenstate $|E\rangle$ with energy E , $\hat{H}|E\rangle = E|E\rangle$, we obtain an eigenstate with energy $-E$, since

$$\hat{H}(\hat{S}|E\rangle) = -(\hat{S} \hat{H} \hat{S}^\dagger) \hat{S}|E\rangle = -\hat{S} \hat{H}|E\rangle = -E(\hat{S}|E\rangle). \quad (2)$$

Therefore the spectrum of $\hat{H}$ is symmetric around $E = 0$.

This quantum wire is a prototypical system that displays different quantum phases upon changes in the parameters of the Hamiltonian that are distinguished by their topological properties. This can be seen more easily if we rotate the spin basis to a direction orthogonal

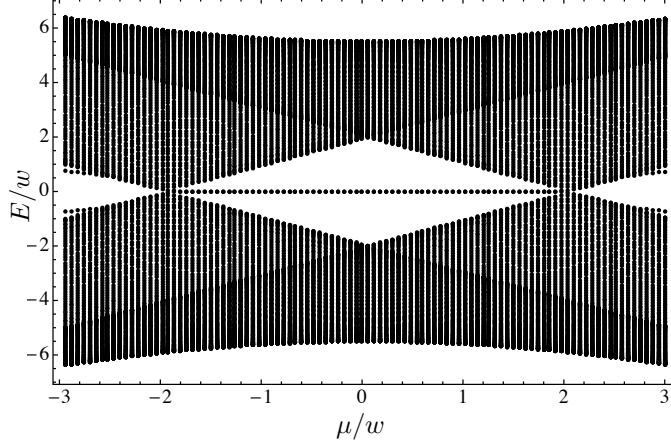


FIG. 1. The single-particle energy spectrum of the quantum wire shows the quantum phases of the system. Here, $N = 150$ and $\lambda/w = 5.5$. In the topological insulator phase, $|\mu/2w| < 1$, a pair of zero-energy bound states is found localized at the two edges of the lattice. They split away in the normal insulator phase.

to the magnetic field. In the new basis, the annihilation operators are

$$\hat{b}_{\uparrow r} = \frac{\hat{c}_{\uparrow r} + \hat{c}_{\downarrow r}}{\sqrt{2}}, \quad (3)$$

$$\hat{b}_{\downarrow r} = \frac{\hat{c}_{\uparrow r} - \hat{c}_{\downarrow r}}{\sqrt{2}}. \quad (4)$$

After some straightforward algebra, we obtain the Hamiltonian in the new basis as

$$\hat{H} = \sum_{r=1}^N \left[\mu \hat{b}_{\uparrow r}^\dagger \hat{b}_{\downarrow r} + \left(w + \frac{\lambda}{2} \right) \hat{b}_{\uparrow r+1}^\dagger \hat{b}_{\downarrow r} + \left(w - \frac{\lambda}{2} \right) \hat{b}_{\downarrow r+1}^\dagger \hat{b}_{\uparrow r} \right] + \text{h.c.} \quad (5)$$

Now we can easily see that when the parameters are at the “sweet spot” $\mu = 0$ and $w = \lambda/2$, the Hamiltonian reduces to $\hat{H} = \lambda \sum_{r=1}^{N-1} \hat{b}_{\uparrow r+1}^\dagger \hat{b}_{\downarrow r} + \text{h.c.}$ Therefore, the two operators $\hat{b}_{\uparrow 1}^\dagger$ and $\hat{b}_{\downarrow N}^\dagger$ (and their Hermitian conjugates) do not appear in the Hamiltonian and $[\hat{H}, \hat{b}_{\uparrow 1}] = [\hat{H}, \hat{b}_{\downarrow N}] = 0$. This means that the bound states created by $\hat{b}_{\downarrow N}^\dagger$ and $\hat{b}_{\uparrow 1}^\dagger$ at the two ends of the wire are degenerate eigenstates of the Hamiltonian (with zero energy).

One can show that even away from the sweet spot, there are two bound eigenstates of the Hamiltonian as long as $|\mu| < 2w$. The energy of these bound states is exponentially small in the system size, $E_b \sim e^{-Na/\xi}$. Here a is the lattice spacing between nearest-neighbor ions and ξ is a length scale that depends on system parameters. In the thermodynamic limit, $N \rightarrow \infty$, we recover two asymptotically exact zero-energy bound states. The existence or

absence of these bound states marks the topological or trivial phases of the system. The existence of these phases is illustrated in Fig. 1, which shows the single-particle energy spectrum of the lattice Hamiltonian as μ is varied.

Instead of studying the phases on the lattice, we focus on the thermodynamic limit in the following and derive a continuum Hamiltonian. This approach has the advantage of simplifying the solutions since it is often easier to solve continuous differential equations rather than discrete difference equations. Also, many different lattice models can have the same continuum description in the thermodynamic limit. Thus, by keeping only the relevant long-distance, universal features of the system in the continuum limit, we will gain a more general perspective on the potential phases of the system.

IV. THE CONTINUUM LIMIT

We shall take the continuum limit of the lattice model as the lattice constant $a \rightarrow 0$ while the length $L = Na$ is fixed. We leave open the choice of whether the length is finite, semi-infinite, or infinite. In this limit, the discrete site index r is turned into a continuous position x . A simple way to deduce the relation between fermion operators $\hat{c}_{sr}$ in the lattice and the field operator $\hat{\Psi}_s(x)$ in the continuum is to demand the proper anti-commutation relations

$$\hat{\Psi}_s(x)\hat{\Psi}_{s'}(x') + \hat{\Psi}_{s'}(x')\hat{\Psi}_s(x) = 0, \quad (6a)$$

$$\hat{\Psi}_s(x)\hat{\Psi}_{s'}^\dagger(x') + \hat{\Psi}_{s'}^\dagger(x')\hat{\Psi}_s(x) = \delta_{ss'}\delta(x - x'), \quad (6b)$$

where $\delta(x - x')$ is the Dirac delta function.¹ Thus, the fermion field $\hat{\Psi}_s(x)$ must have the dimension of $(\text{length})^{-1/2}$. Defining $x = ar$, $x' = ar'$, and $\lim_{a \rightarrow 0}(\delta_{rr'}/a) = \delta(x - x')$, we see that the proper definition of continuum operators has the form $\hat{\Psi}_s(x) = \lim_{a \rightarrow 0}(\hat{c}_{sr}/\sqrt{a})$. In the limit $a \rightarrow 0$ we can expand

$$\frac{1}{\sqrt{a}}\hat{c}_{sr+1} \rightarrow \hat{\Psi}_s(x + a) = \hat{\Psi}_s(x) + a\frac{d}{dx}\hat{\Psi}_s(x) + O(a^2). \quad (7)$$

Also in this limit $a\sum_r \rightarrow \int dx$. Replacing these limits in the Hamiltonian, Eq. (1), we find several terms that, to first order in a , vanish upon integration. In particular, terms of the form

$$aw \int \hat{\Psi}_s^\dagger(x)\frac{d}{dx}\hat{\Psi}_s(x) dx + \text{h.c.} = aw \int \frac{d}{dx} \left[\hat{\Psi}_s^\dagger(x)\hat{\Psi}_s(x) \right] dx \quad (8)$$

vanish in the limit $a \rightarrow 0$ for finite w . After some straightforward algebra, we find the continuum Hamiltonian

$$\hat{H} \rightarrow \hat{H}_c = \int \hat{\Psi}^\dagger(x) \mathcal{H} \hat{\Psi}(x) dx, \quad (9)$$

where

$$\hat{\Psi}(x) = \begin{pmatrix} \hat{\Psi}_\uparrow(x) \\ \hat{\Psi}_\downarrow(x) \end{pmatrix}, \quad \mathcal{H} = \begin{pmatrix} m & v \frac{d}{dx} \\ -v \frac{d}{dx} & -m \end{pmatrix}, \quad (10)$$

with $m = 2w + \mu$ and $v = a\lambda$. Note that in this limit we keep v finite. Using the Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (11)$$

the matrix $\mathcal{H}$ could be written in a compact form as

$$\mathcal{H} = iv\sigma_y \frac{d}{dx} + m\sigma_z. \quad (12)$$

We note that this is the Hamiltonian leading to the one-dimensional Dirac equation.¹¹ The spin-flip operation of the lattice Hamiltonian, $\hat{S}$ of Sec. III, is now implemented by the Pauli matrix $\mathcal{S} = \sigma_x$ and $\mathcal{S}\mathcal{H}\mathcal{S}^\dagger = -\mathcal{H}$.

The matrix $\mathcal{H}$ is an operator in the position basis. Its eigenvalues E and eigenstates¹² $\phi_E(x) = (\phi_{\uparrow E}(x) \ \phi_{\downarrow E}(x))^\top$ are found by solving the linear system of differential equations

$$\mathcal{H}\phi_E(x) = E\phi_E(x). \quad (13)$$

Using these solutions, we can define the energy creation operators

$$\hat{\Psi}_E^\dagger = \int \left[\phi_{\uparrow E}(x) \hat{\Psi}_\uparrow^\dagger(x) + \phi_{\downarrow E}(x) \hat{\Psi}_\downarrow^\dagger(x) \right] dx. \quad (14)$$

As we show in Appendix A, the $\hat{\Psi}_E^\dagger$ are “ladder operators” that satisfy the commutation relations $[\hat{H}_c, \hat{\Psi}_E^\dagger] = E\hat{\Psi}_E^\dagger$. In this way, the full spectrum of the continuum Hamiltonian can be obtained by solving the first-quantized eigenvalue equation (13). We will concentrate on studying the properties of $\mathcal{H}$ in the rest of this paper.

The careful reader may have noticed that the energy E in Eq. (13) can be unboundedly large. For instance, consider a solution such as

$$\phi_E(x) = e^{ikx} \begin{pmatrix} 1 \\ \frac{E-m}{ivk} \end{pmatrix}, \quad (15)$$

with eigenvalue $E = \pm\sqrt{v^2k^2 + m^2}$. The energy E can be an indefinitely large number for large k . Thus $\mathcal{H}$ and the continuum Hamiltonian $\hat{H}_c$, unlike the lattice Hamiltonian, are unbounded operators. More surprisingly, even though the Hamiltonian is formally Hermitian, in a finite geometry ($x < \infty$ and/or $x > -\infty$) the energy eigenvalues can be imaginary numbers. We can see this by replacing k with $i\kappa$ in the above solution. Such a solution would be acceptable since $e^{-\kappa x}$ need not diverge in a finite geometry for a proper choice of $\text{sgn}(\kappa)$. When $|v\kappa| > |m|$, the energy is a purely imaginary number.

The presence of imaginary eigenvalues is problematic for the correspondence of the Hamiltonian to observable energy. Formally, it signals the loss of self-adjointness of the Hamiltonian operator. To restore self-adjointness, the Hamiltonian needs to be properly defined on a subset of the Hilbert space such that no imaginary eigenvalues exist in the spectrum. In the following two sections we will introduce the mathematical technique to deal with this problem for a general unbounded operator, and find the appropriate subset that resolves this issue for our continuum Hamiltonian.

The following exercise shows that there is a second continuum field operator that one can define as $\hat{\Phi}_s(x) = \lim_{a \rightarrow 0} (-1)^r \hat{c}_{sr} / \sqrt{a}$.

Exercise 1. By expanding the field operators $\hat{\Phi}_s$ in the limit $a \rightarrow 0$, show that the continuum Hamiltonian for $\hat{\Phi} = (\hat{\Phi}_\uparrow \ \hat{\Phi}_\downarrow)^\top$ is

$$\hat{H} \rightarrow \int \hat{\Phi}^\dagger(x) \left[-iv\sigma_y \frac{d}{dx} + (\mu - 2w)\sigma_z \right] \hat{\Phi}(x) dx. \quad (16)$$

Thus, as the analysis in the following sections shows, the continuum limit in terms of $\hat{\Phi}$ becomes important for the topological phases of the system when $\mu - 2w$ changes sign. For simplicity, we shall assume $2w > \mu$ in the following.

V. THE SELF-ADJOINT EXTENSIONS OF A HERMITIAN OPERATOR

The state of a quantum system is specified by a vector ϕ in a Hilbert space $\mathcal{H}$.¹ The Hilbert space is equipped with an inner product $\langle \psi | \phi \rangle$ that is a complex-valued, positive-definite, sesquilinear function on $\mathcal{H} \times \mathcal{H}$ that maps any two states $\phi, \psi \in \mathcal{H}$ to a complex number $\langle \psi | \phi \rangle = \langle \phi | \psi \rangle^*$. The inner product defines a norm for the Hilbert space, $\|\phi\| \equiv \sqrt{\langle \phi | \phi \rangle}$. For example, given an interval $[L_1, L_2] \subset \mathbb{R}$, the Hilbert space $\mathcal{H} = \mathcal{L}^2[L_1, L_2]$ is

the set of all square-integrable functions with the inner-product $\langle \psi | \phi \rangle = \int_{L_1}^{L_2} \psi^*(x) \phi(x) dx$, i.e., those functions with $\|\phi\| < \infty$.

Physical observables of a quantum system are specified by linear self-adjoint operators.¹ For an operator A to be self-adjoint, it primarily must be Hermitian (or symmetric). An operator A is defined to be Hermitian if

$$\langle \psi | A \phi \rangle = \langle A \psi | \phi \rangle \quad (17)$$

for all vectors ψ and ϕ in its domain. In many textbooks on quantum mechanics, no distinction is made between Hermitian and self-adjoint operators. Indeed, one need not make such a distinction for bounded operators. (A bounded operator B is one for which there exists a positive number M such that $\|B\phi\| < M\|\phi\|$ for all vectors ϕ in the Hilbert space.) However, the distinction between Hermitian and self-adjoint operators becomes crucial for unbounded operators. This is so because for unbounded operators the designation of the domain of the operator A , $\mathcal{D}(A)$, which is a subspace of the full Hilbert space $\mathcal{H}$, $\mathcal{D}(A) \subset \mathcal{H}$, generally becomes an important part of its definition.

For Hermitian operators, the connection between unboundedness and a restricted domain is quite general. In fact, according to the Hellinger-Toeplitz theorem,¹³ a Hermitian operator B with the domain $\mathcal{D}(B) = \mathcal{H}$ is bounded. It is a direct corollary of this theorem that an unbounded and Hermitian operator cannot be defined on the whole Hilbert space. Let us illustrate this point by an example. Consider the position operator, X , on the Hilbert space $\mathcal{L}^2[0, L]$ defined as $(X\phi)(x) = x\phi(x)$ for $\phi \in \mathcal{L}^2[0, L]$. Equation (17) is satisfied since

$$\langle \psi | X \phi \rangle = \int_0^L \psi^*(x) x \phi(x) dx = \int_0^L (x \psi(x))^* \phi(x) dx = \langle X \psi | \phi \rangle. \quad (18)$$

Thus, the position operator is a Hermitian operator. In this case, X is a bounded operator since for any $\phi \in \mathcal{L}^2[0, L]$, $\|X\phi\| < M\|\phi\|$ with $M = L$. Now, if instead we consider the whole real line, $\mathbb{R}$, the position operator defined on the Hilbert space $\mathcal{L}^2(\mathbb{R})$ continues to be Hermitian, but is no longer bounded. Now, consider $\phi \in \mathcal{L}^2(\mathbb{R})$ such that $\phi(x) = x/(x^2+1)$. It is clear that $(X\phi)(x) = x^2/(x^2+1)$ is not square-integrable and therefore $X\phi$ is ill-defined. Therefore, in this case X cannot be defined on the whole Hilbert space and one must find an appropriate domain for X , such as $\mathcal{D}(X) = \{\phi \in \mathcal{L}^2(\mathbb{R}) : \|X\phi\| < \infty\}$. This means $\phi(x) \rightarrow 0$ for $x \rightarrow \infty$ faster than $x^{-3/2}$. As we shall discuss in detail below, restrictions on the domain of an operator are commonly expressed in terms of boundary conditions on the state vectors.

In order to define a self-adjoint operator properly, we first need to define the adjoint of an operator and its domain. For an operator $A : \mathcal{D}(A) \rightarrow \mathcal{H}$ and $\phi \in \mathcal{D}(A)$, the adjoint $A^\dagger$ and its domain of states $\psi \in \mathcal{D}(A^\dagger)$ are defined by

$$\langle \phi | A^\dagger \psi \rangle := \langle A \phi | \psi \rangle. \quad (19)$$

(This defines $A^\dagger \psi$ uniquely only if the operator A is “densely defined;” see Appendix B for a discussion.) A self-adjoint operator A is then defined as one that is Hermitian *and* has the same domain as its adjoint, $\mathcal{D}(A^\dagger) = \mathcal{D}(A)$.

As an example, consider the momentum operator, P , on the Hilbert space $\mathcal{L}^2[0, L]$, defined as $(P\phi)(x) = -i\hbar d\phi/dx$ (we use units such that Planck’s constant $\hbar = 1$). Since the derivative of a square-integrable function can be arbitrarily large, P is unbounded. For its domain we take $\mathcal{D}(P) = \{\phi \in \mathcal{L}^2[0, L] : \phi(x) \text{ is absolutely continuous and } \phi(0) = \phi(L) = 0\}$.¹⁴ (One can show that with this definition, P is densely defined; see Appendix B.) Simple algebra shows that Eq. (19) holds true if we have

$$(P^\dagger \psi)(x) = -i \frac{d}{dx} \psi(x), \quad (20a)$$

$$\psi^*(L)\phi(L) = \psi^*(0)\phi(0). \quad (20b)$$

The first equation shows that the momentum operator is a Hermitian operator. However, the second equation shows that $\psi(0)$ and $\psi(L)$ could be nonzero and still Eq. (19) would be satisfied. That is, $\mathcal{D}(P^\dagger) = \{\psi \in \mathcal{L}^2[0, L] : \psi(x) \text{ is absolutely continuous}\} \supset \mathcal{D}(P)$. Therefore, P is not self-adjoint. Some puzzling consequences of the lack of self-adjointness for P are discussed in Ref. 6. An important consequence for our purposes is that the domain of the adjoint operator $P^\dagger$ is so unconstrained that it admits *imaginary* eigenvalues. That is, one can find states $\psi_\pm \in \mathcal{D}(P^\dagger)$ such that $P^\dagger \psi_\pm = \pm i\eta \psi_\pm$ for $\eta > 0$. Indeed, one can easily show that the two states

$$\psi_+(x) = e^{-\eta x}, \quad \psi_-(x) = e^{-\eta(L-x)}, \quad (21)$$

are such eigenstates with equal norm, $\|\psi_+\| = \|\psi_-\|$.¹⁵

As seen in this example, the domain of the Hermitian operator A is no larger than the domain of its adjoint, $A^\dagger$. This means that in order to make the operator A self-adjoint one must *extend* its domain in a way that shrinks the domain of the adjoint to match it, and eliminate the imaginary eigenvalues from the spectrum of the adjoint operator. If the

procedure is successful, we find a self-adjoint extension of the Hermitian operator A . This procedure was made systematic by von Neumann by a method known as the deficiency indices method.

In this method, given a Hermitian, densely defined, and closed operator $A : \mathcal{D}(A) \rightarrow \mathcal{H}$ (see Appendix B for the definitions), one first finds the adjoint operator $A^\dagger$ and its domain $\mathcal{D}(A^\dagger)$ and then solves for the imaginary eigenstates of the adjoint operator. These solutions define the two *deficiency* subspaces $\mathcal{K}_\pm = \{\psi_\pm \in \mathcal{D}(A^\dagger), A^\dagger \psi_\pm = \pm i\eta \psi_\pm\}$. The dimensions of these subspaces, $n_\pm = \dim(\mathcal{K}_\pm)$, are the deficiency indices of A .

Deficiency indices theorem (Von Neumann).^{5,13,16} For a Hermitian, densely defined, and closed operator A ,

1. If $n_+ = n_- = 0$, A is self-adjoint.
2. If $n_+ \neq n_-$, A is not self-adjoint and it has no self-adjoint extensions.
3. If $n_+ = n_- = n > 0$, A has infinitely many self-adjoint extensions $A_\mathcal{U}$ parametrized by $n \times n$ unitary matrix maps $\mathcal{U} : \mathcal{K}_+ \rightarrow \mathcal{K}_-$. For a choice of $\mathcal{U}$, $\mathcal{D}(A) \subset \mathcal{D}(A_\mathcal{U}) \subset \mathcal{D}(A^\dagger)$,

$$\mathcal{D}(A_\mathcal{U}) = \{\psi = \phi + \psi_+ + \mathcal{U}\psi_+ : \phi \in \mathcal{D}(A), \psi_+ \in \mathcal{K}_+\}, \quad (22)$$

$$A_\mathcal{U}\psi = A^\dagger\psi = A\phi + i\eta\psi_+ - i\eta\mathcal{U}\psi_+. \quad (23)$$

Note that even though η appears explicitly in the deficiency subspaces, the family of self-adjoint extensions is independent of the parameter η .⁶ So, sometimes the choice $\eta = 1$ is made. However, we will keep η for dimensional purposes.

Considering the example of the momentum operator P on $\mathcal{L}^2[0, L]$, we see that $n_+ = n_- = 1$. Therefore there is a one-parameter family of self-adjoint extensions $P_\mathcal{U} \equiv P_\theta$ with $\mathcal{U} : \psi_+ \mapsto e^{i\theta}\psi_-$ and $\psi_\pm$ given in Eq. (21), such that

$$P_\theta\psi = -i\frac{d}{dx}\psi, \quad (24)$$

$$\mathcal{D}(P_\theta) = \{\psi = \phi + c(\psi_+ + e^{i\theta}\psi_-) : \phi(0) = \phi(L) = 0, c \in \mathbb{C}\}. \quad (25)$$

Note that $\psi \in \mathcal{D}(P_\theta)$ satisfies a new boundary condition, $\psi(L) = e^{i\beta}\psi(0)$, with $e^{i\beta} = (e^{-\eta L} + e^{i\theta})/(1 + e^{-\eta L}e^{i\theta})$.

While this method is systematic, it is rather abstract. In Sec. VII we will introduce a more intuitive way to find the proper domain of self-adjoint extensions in terms of boundary conditions and the local current associated with the corresponding self-adjoint operator.

VI. SELF-ADJOINT EXTENSIONS AND TOPOLOGICAL BOUND STATES

We will now apply von Neumann's deficiency indices method to obtain self-adjoint extensions of the continuum Hamiltonian $\mathcal{H}$, Eq. (12), in semi-infinite and finite-size wire geometries. In each case, we will show that the appropriately extended domains admit additional bound states in the topological phase, and we will relate their properties to the boundary conditions at the ends of the wire. We will also see that demanding that the spectral symmetry, given by the spin-flip operation $\mathcal{S}$ of Sec. IV, be preserved by the self-adjoint extension forces the bound states to have zero energy.

A. Semi-infinite wire

The simplest geometry with a boundary is the semi-infinite wire, $x \in [0, \infty)$, with just one physical boundary. In order to apply the deficiency theorem, we choose the domain

$$\mathcal{D}(\mathcal{H}) = \left\{ \phi = \begin{pmatrix} \phi_{\uparrow} \\ \phi_{\downarrow} \end{pmatrix} : \phi_s(x) \in \mathcal{L}^2[0, \infty), \phi_s(0) = 0 \right\}, \quad (26)$$

where s is the spin. (This domain is dense; the proof is similar to the example given in Appendix B). With this choice, the adjoint operator has the domain $\mathcal{D}(\mathcal{H}^\dagger) = \{\psi : \psi_s(x) \in \mathcal{L}^2[0, \infty)\}$. Now we can find all possible extensions $\mathcal{H}_{\mathcal{U}}$ such that $\mathcal{D}(\mathcal{H}_{\mathcal{U}}) = \mathcal{D}(\mathcal{H}_{\mathcal{U}}^\dagger)$.

Next, we solve for $\mathcal{H}^\dagger \psi_{\pm} = \pm i\eta \psi_{\pm}$. It is easy to see that

$$\psi_{\pm}(x) = e^{-|\epsilon|x/v} \begin{pmatrix} 1 \\ e^{\pm i\delta} \end{pmatrix}, \quad (27)$$

where $|\epsilon|e^{i\delta} = m - i\eta$. (Notice that $\delta \neq 0, \pi$ and $\|\psi_+\| = \|\psi_-\|$.) So the deficiency indices are $n_+ = n_- = n = 1$. The domain of the extended operator $\mathcal{H}_{\mathcal{U}} \equiv \mathcal{H}_{\theta}$ is parametrized by the one-parameter mapping $\mathcal{U} : \psi_+ \mapsto e^{i\theta} \psi_-$ as

$$\mathcal{D}(\mathcal{H}_{\theta}) = \{\psi = \phi + c(\psi_+ + e^{i\theta} \psi_-) : \phi(0) = 0, c \in \mathbb{C}\}. \quad (28)$$

The domain $\mathcal{D}(\mathcal{H}_\theta)$ is equivalently characterized by a new boundary condition. To see this, note that for an arbitrary state $\psi \in \mathcal{D}(\mathcal{H}_\theta)$ we have

$$\psi(0) = \begin{pmatrix} \psi_\uparrow(0) \\ \psi_\downarrow(0) \end{pmatrix} = c \begin{pmatrix} 1 + e^{i\theta} \\ e^{i\delta} + e^{i\theta} e^{-i\delta} \end{pmatrix}. \quad (29)$$

Thus

$$\psi_\uparrow(0) = \Lambda \psi_\downarrow(0), \quad (30a)$$

$$\Lambda = \frac{\cos(\theta/2)}{\cos(\theta/2 - \delta)} \in \mathbb{R}. \quad (30b)$$

Since we have extended the domain of $\mathcal{H}$, we may expect the spectrum of $\mathcal{H}_\theta$ to depend on the choice of θ or, equivalently, Λ . Indeed, one can easily check that for $m\Lambda > 0$ there is always a bound state,

$$\psi_b(x) = \exp\left(-\frac{m}{v} \frac{2\Lambda}{\Lambda^2 + 1} x\right) \begin{pmatrix} \Lambda \\ 1 \end{pmatrix}, \quad (31)$$

with energy eigenvalue

$$E_b = \frac{\Lambda^2 - 1}{\Lambda^2 + 1} m. \quad (32)$$

Note that $-|m| < E_b < |m|$, i.e., it is within the energy gap of the bulk states.

As the careful reader may infer from the dependence of the spectrum on the choice of self-adjoint extension, other physical properties may also depend on this choice. An important case is the effect of a symmetry operation. For example, what happens to the spectral symmetry given by the spin-flip operation $\mathcal{S} = \sigma_x$ of Sec. IV? It is easy to see that under the operation $\mathcal{S}$ the boundary condition characterizing $\mathcal{D}(\mathcal{H}_\theta)$ is mapped as $\Lambda \mapsto \Lambda' = 1/\Lambda$. Therefore, $\mathcal{S}$ is not in general a symmetry of the spectrum and the spectrum may not be symmetric around zero. Indeed, the existence of a single bound state with $E_b \neq 0$ is possible only if the spectrum is not symmetric. The spectral symmetry is restored for $\Lambda' = \Lambda = \pm 1$; for $\Lambda = \text{sgn}(m)$ the spectrum has a zero-energy bound state, $E_b = 0$. Therefore, whether or not a symmetry of the bulk lattice is a symmetry of the continuum description depends on the choice of self-adjoint extension.

B. Finite wire

We now consider a more realistic wire geometry: a wire with a finite length L . We choose the domain of the Hamiltonian in Eq. (12) to be

$$\mathcal{D}(\mathcal{H}) = \{\phi : \phi_s \in \mathcal{L}^2[0, L], \phi_s(0) = \phi_s(L) = 0\}, \quad (33)$$

where s is the spin. (One can show that this is a dense domain similar to previous cases.) With this choice, the domain of the adjoint operator is

$$\mathcal{D}(\mathcal{H}^\dagger) = \{\psi : \psi_s \in \mathcal{L}^2[0, L]\}. \quad (34)$$

The deficiency indices are determined by solving the imaginary eigenvalue equation $\mathcal{H}^\dagger \psi_\pm = \pm i\eta \psi_\pm$. We now find *two* independent solutions for each sign,

$$\psi_\pm^{(1)}(x) = e^{-|\epsilon|x/v} \begin{pmatrix} 1 \\ e^{\pm i\delta} \end{pmatrix}, \quad \psi_\pm^{(2)}(x) = e^{-|\epsilon|(L-x)/v} \begin{pmatrix} 1 \\ -e^{\pm i\delta} \end{pmatrix}, \quad (35)$$

with $|\epsilon|e^{i\delta} = m - i\eta$ as before. (Note also that $\|\psi_+^{(1)}\| = \|\psi_-^{(1)}\| = \|\psi_+^{(2)}\| = \|\psi_-^{(2)}\|$.) The deficiency spaces $\mathcal{K}_\pm = \{\psi_\pm = c_\pm^{(1)}\psi_\pm^{(1)} + c_\pm^{(2)}\psi_\pm^{(2)} : c_\pm^{(1)}, c_\pm^{(2)} \in \mathbb{C}\}$ are two dimensional; the deficiency indices are $n_+ = n_- = n = 2$. Consequently, the extended domain is parametrized by 2×2 unitary matrix maps

$$\mathcal{U} : \begin{pmatrix} \psi_+^{(1)} \\ \psi_+^{(2)} \end{pmatrix} \mapsto \begin{pmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{pmatrix} \begin{pmatrix} \psi_-^{(1)} \\ \psi_-^{(2)} \end{pmatrix} \equiv u \begin{pmatrix} \psi_-^{(1)} \\ \psi_-^{(2)} \end{pmatrix}. \quad (36)$$

The domain of the self-adjoint extension is

$$\mathcal{D}(\mathcal{H}_\mathcal{U}) = \{\psi = \phi + \psi_+ + \mathcal{U}\psi_+ : \phi_s(0) = \phi_s(L) = 0, \psi_+ \in \mathcal{K}_+\}. \quad (37)$$

What boundary conditions characterize $\mathcal{D}(\mathcal{H}_\mathcal{U})$? To find the answer, we note that for any $\chi, \psi \in \mathcal{D}(\mathcal{H}_\mathcal{U})$, we must have

$$\langle \chi | \mathcal{H}_\mathcal{U} \psi \rangle = \langle \mathcal{H}_\mathcal{U} \chi | \psi \rangle. \quad (38)$$

In particular, this must be true if we choose $\chi = \psi_+^{(1)} + \mathcal{U}\psi_+^{(1)}$ and $\chi = \psi_+^{(2)} + \mathcal{U}\psi_+^{(2)}$ separately. By partial integration, it is easy to see that

$$\langle \psi_\pm^{(a)} | \mathcal{H}_\mathcal{U} \psi \rangle = \langle \mathcal{H}_\mathcal{U} \psi_\pm^{(a)} | \psi \rangle + iv \left[\psi_\pm^{(a)}(x)^\dagger \sigma_y \psi(x) \right]_0^L, \quad a = 1, 2. \quad (39)$$

The boundary terms are straightforward to calculate and yield

$$\begin{aligned}
i \left[\psi_{\pm}^{(a)}(x)^{\dagger} \sigma_y \psi(x) \right]_0^L &= \exp \left(\frac{\zeta_a - 1}{2} \frac{|\epsilon|}{v} L \right) [\psi_{\downarrow}(L) + \zeta_a e^{\mp i\delta} \psi_{\uparrow}(L)] \\
&\quad - \exp \left(-\frac{\zeta_a + 1}{2} \frac{|\epsilon|}{v} L \right) [\psi_{\downarrow}(0) + \zeta_a e^{\mp i\delta} \psi_{\uparrow}(0)] \\
&\equiv \pm \xi_{\pm}^{(a)},
\end{aligned} \tag{40}$$

with $\zeta_a = (-1)^a$. From Eq. (38), collecting the boundary terms, we find the matrix equation $u \xi_+[\psi] = \xi_-[\psi]$, where $\xi_+[\psi] = (\xi_+^{(1)} \ \xi_+^{(2)})^{\top}$ and $\xi_-[\psi] = (\xi_-^{(1)} \ \xi_-^{(2)})^{\top}$. Since we know that the family of self-adjoint extensions is the same for all choices of η , we can rewrite this for a particular choice of η ; namely we shall take the limiting case $\eta \rightarrow \infty$ (i.e., $|\epsilon| \rightarrow \infty$ and $e^{-i\delta} \rightarrow i$) to find

$$u \begin{pmatrix} -\psi_{\downarrow}(0) + i\psi_{\uparrow}(0) \\ \psi_{\downarrow}(L) + i\psi_{\uparrow}(L) \end{pmatrix} = \begin{pmatrix} \psi_{\downarrow}(0) + i\psi_{\uparrow}(0) \\ -\psi_{\downarrow}(L) + i\psi_{\uparrow}(L) \end{pmatrix}. \tag{41}$$

The boundary condition can also be written in a form that appears independent of $\mathcal{U}$. This can be done by noting that the boundary conditions relate the two vectors ξ_+ and ξ_- by a unitary matrix u . Equivalently, for two states $\psi, \phi \in \mathcal{D}(\mathcal{H}_{\mathcal{U}})$ the inner product between the two vectors $\xi_{\pm}[\psi]$ and $\xi_{\pm}[\phi]$ defined separately for ψ and ϕ must be preserved, i.e., $\xi_+[\psi]^{\dagger} \xi_+[\phi] = \xi_-[\psi]^{\dagger} \xi_-[\phi]$. After some algebra this yields

$$\psi_{\uparrow}^*(L) \phi_{\downarrow}(L) - \psi_{\downarrow}^*(L) \phi_{\uparrow}(L) = \psi_{\uparrow}^*(0) \phi_{\downarrow}(0) - \psi_{\downarrow}^*(0) \phi_{\uparrow}(0). \tag{42}$$

This condition is quadratic in the wavefunctions and so is not immediately useful in solving the differential equations needed to find the spectrum. However, it is equivalent to the linear boundary conditions in Eq. (41). We note that it reduces to the boundary condition we found for the semi-infinite geometry if we let $L \rightarrow \infty$ and set $\psi_s(\infty) = 0$.

We next study the spectral symmetry as in the previous case. For the spin-flip operation, $\mathcal{S}$, we find $\xi_{\pm}[\mathcal{S}\psi] = \pm i\sigma_z \xi_{\mp}[\psi]$. Thus, using the fact that u is unitary, the boundary condition is mapped to $u' \xi_+[\psi] = \xi_-[\psi]$ with $u' = -\sigma_z u^{\dagger} \sigma_z$. A boundary condition preserving the spectral symmetry is found when $u = u'$. This latter condition can be rewritten as $\mathcal{M} = \mathcal{M}^{\dagger} = \mathcal{M}^{-1}$ for $\mathcal{M} = i\sigma_z u$. That is, $\mathcal{M}$ is Hermitian and squares to $\mathbf{1}$. There are two types of solutions to this equation: (1) $\mathcal{M} = \pm \mathbf{1}$, yielding $u = \mp i\sigma_z$; and (2) $\mathcal{M} = m_x \sigma_x + m_y \sigma_y + m_z \sigma_z \equiv \mathbf{m} \cdot \boldsymbol{\sigma}$, with a unit vector $\mathbf{m} = (m_x, m_y, m_z) \in S^2$ on the real two-sphere, yielding $u = -i\sigma_z(\mathbf{m} \cdot \boldsymbol{\sigma})$. It is not difficult to show that in the first case, the

boundary conditions for the wavefunctions read

$$\psi_{\uparrow}(0) = \pm\psi_{\downarrow}(0), \quad \psi_{\uparrow}(L) = \pm\psi_{\downarrow}(L). \quad (43)$$

As an example of the second case, we take $u = \mp i\mathbf{1}$. Then,

$$\psi_{\uparrow}(0) = \pm\psi_{\downarrow}(0), \quad \psi_{\uparrow}(L) = \mp\psi_{\downarrow}(L). \quad (44)$$

The effect on the spectrum can be seen directly in terms of the number of zero-energy states in each case. With the boundary conditions in Eq. (43) there is always *one* exact zero-energy state,

$$\psi_0^{(+)}(x) = e^{-mx/v} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{or} \quad \psi_0^{(-)}(x) = e^{-m(L-x)/v} \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \quad (45)$$

For the the boundary conditions in Eq. (44) there are no exact zero-energy states for finite L . However, for large L the conditions with the upper sign are satisfied by both $\psi_0^{(+)}$ and $\psi_0^{(-)}$ to the order $e^{-mL/v}$, while for the lower sign the conditions are never satisfied independent of L . Thus, in this case, there are no zero-energy states for the lower sign and there are *two* asymptotically exact zero-energy states for the upper sign.

Thus, we see that even after imposing the spectral symmetry given by $\mathcal{S}$, the bulk-boundary correspondence may still be incomplete. This can be understood in terms of the physical implications of the boundary conditions in Eqs. (43) and (44). These conditions can be understood as modeling the interface of the wire with an environment. Note that the condition $\psi_{\uparrow}(0) = \psi_{\downarrow}(0)$ in the semi-infinite case allowed the existence of a zero-energy state localized at the edge, while the condition $\psi_{\uparrow}(0) = -\psi_{\downarrow}(0)$ did not. Therefore, the former condition models the interface of a normal insulator, the “vacuum,” with the *left* edge of the wire that supports a topological bound state. If we had chosen the semi-infinite geometry $x < 0$ with a *right* edge, it is easy to see that the sign of the boundary condition modeling this situation would have been the opposite. This shows up in the finite geometry: now the boundary conditions in Eqs. (43) and (44) at the two edges separately model an interface with vacuum with the $+$ sign on the left edge and the $-$ sign at the right edge.

VII. SELF-ADJOINT EXTENSIONS AND CONSERVED CURRENTS

As we have seen in several examples, such as in Eq. (42), the conditions determining the domain of self-adjoint extensions of a Hermitian operator were written in a way that was

independent of the specific choice of $\mathcal{U}$. In this section we show that this is quite a general result. Furthermore, we show that this can be directly interpreted as the conservation of a local current associated with the self-adjoint operator.

Given a self-adjoint operator A and $\psi, \phi \in \mathcal{D}(A)$, we must have $\langle \psi | A \phi \rangle - \langle A \psi | \phi \rangle = 0$. This equation is the basis of the self-adjoint extensions we have so far discussed. For example, for the momentum operator $P = -id/dx$ on the Hilbert space $\mathcal{L}^2[0, L]$ we have

$$\begin{aligned} 0 &= \langle \psi | P \phi \rangle - \langle P \psi | \phi \rangle \\ &= \int_0^L \left[\psi^*(x) \left(-i \frac{d}{dx} \phi(x) \right) - \left(-i \frac{d}{dx} \psi(x) \right)^* \phi(x) \right] dx \\ &= -i \int_0^L \frac{d}{dx} (\psi^*(x) \phi(x)) dx. \end{aligned} \quad (46)$$

As a result, we find $\psi^*(L)\phi(L) - \psi^*(0)\phi(0) = 0$. The general solution independent of ψ and ϕ is $\psi(L) = e^{i\beta}\psi(0)$ as we found under Eq. (25). Note that for $\psi = \phi$ this condition reads

$$|\psi(L)|^2 = |\psi(0)|^2, \quad (47)$$

which is simply the conservation of probability at the two boundaries. Since the momentum operator generates space translations, the probability density can be interpreted as the conserved current under space translations. Therefore, this condition simply states that the probability density must be conserved throughout the interval $[0, L]$.

Indeed, we show now that this way of specifying self-adjoint extensions of a Hermitian operator is quite general. For a self-adjoint operator A one can always define a one-parameter family of unitary operators $U_A(\alpha) = e^{-i\alpha A}$, generated by A . This result is known as Stone's theorem.¹³ For example, the Hamiltonian generates the time-evolution operator with α being the time parameter. Then we can “evolve” any given state $\psi \in \mathcal{D}(A)$ as $\psi(\alpha) = U_A(\alpha)\psi$ with the initial condition $\psi(0) = \psi$. Since for two such evolved states, $\psi(\alpha)$ and $\phi(\alpha)$, the inner product $\langle \psi(\alpha) | \phi(\alpha) \rangle = \int \psi(x, \alpha)^* \phi(x, \alpha) dx$ is independent of α , the density $\rho_A(x, \alpha) \equiv \psi(x, \alpha)^* \phi(x, \alpha)$ must have a *local* conserved current, $j_A(x, \alpha)$, associated with it such that the following continuity equation is satisfied:

$$\frac{\partial}{\partial \alpha} \rho_A + \frac{\partial}{\partial x} j_A = 0. \quad (48)$$

To see this, note that since $id\psi(\alpha)/d\alpha = A\psi(\alpha)$, we have

$$i \frac{\partial}{\partial \alpha} \rho_A(x, \alpha) = \psi^*(x, \alpha) (A\phi)(x, \alpha) - (A\psi)^*(x, \alpha) \phi(x, \alpha). \quad (49)$$

Thus, the self-adjoint condition can be directly written as

$$\begin{aligned}
0 &= i[\langle \psi | A \phi \rangle - \langle A \psi | \phi \rangle] \\
&= - \int \frac{\partial}{\partial \alpha} \rho_A(x, \alpha) dx \\
&= \int \frac{\partial}{\partial x} j_A(x, \alpha) dx.
\end{aligned} \tag{50}$$

Since this is satisfied for all α , and in particular $\alpha = 0$, we find that the conserved current

$$j_A(x) = i \int^x [\psi^*(x')(A\phi)(x') - (A\psi)^*(x')\phi(x')] dx' \tag{51}$$

characterizes the domain of the self-adjoint operator A .

For the continuum Hamiltonian $\mathcal{H}$, Eq. (12), one finds the local current

$$j_{\mathcal{H}}(x) = -v\psi(x)^\dagger \sigma_y \phi(x) = iv[\psi_\uparrow^*(x)\phi_\downarrow(x) - \psi_\downarrow^*(x)\phi_\uparrow(x)]. \tag{52}$$

So, in the semi-infinite geometry, noting that $j_{\mathcal{H}}(\infty) = 0$, we find $j_{\mathcal{H}}(0) = 0$. This reads $\psi_\uparrow^*(0)\phi_\downarrow(0) = \psi_\downarrow^*(0)\phi_\uparrow(0)$, which indeed yields Eq. (30) independent of the states. In the finite geometry $x \in [0, L]$ we have $j_{\mathcal{H}}(L) = j_{\mathcal{H}}(0)$, which is precisely the condition we found in Eq. (42). In order to obtain the boundary conditions from the current condition, we must revert the steps that took us from Eq. (41) to Eq. (42). To do so, we use the identity $j_{\mathcal{H}} = \frac{1}{2}v[(-\psi_\downarrow + i\psi_\uparrow)^*(-\phi_\downarrow + i\phi_\uparrow) - (\psi_\downarrow + i\psi_\uparrow)^*(\phi_\downarrow + i\phi_\uparrow)]$ to write $j_{\mathcal{H}}(L) = j_{\mathcal{H}}(0)$ as

$$\begin{pmatrix} -\psi_\downarrow(0) + i\psi_\uparrow(0) \\ \psi_\downarrow(L) + i\psi_\uparrow(L) \end{pmatrix}^\dagger \begin{pmatrix} -\phi_\downarrow(0) + i\phi_\uparrow(0) \\ \phi_\downarrow(L) + i\phi_\uparrow(L) \end{pmatrix} = \begin{pmatrix} \psi_\downarrow(0) + i\psi_\uparrow(0) \\ -\psi_\downarrow(L) + i\psi_\uparrow(L) \end{pmatrix}^\dagger \begin{pmatrix} \phi_\downarrow(0) + i\phi_\uparrow(0) \\ -\phi_\downarrow(L) + i\phi_\uparrow(L) \end{pmatrix}. \tag{53}$$

This means that the boundary conditions must map

$$\begin{pmatrix} -\psi_\downarrow(0) + i\psi_\uparrow(0) \\ \psi_\downarrow(L) + i\psi_\uparrow(L) \end{pmatrix} \mapsto \begin{pmatrix} \psi_\downarrow(0) + i\psi_\uparrow(0) \\ -\psi_\downarrow(L) + i\psi_\uparrow(L) \end{pmatrix}, \tag{54}$$

such that the inner product in Eq. (53) is preserved independently of the state ψ . This is satisfied if and only if the states satisfy the boundary condition in Eq. (41). This method can be used to obtain the boundary conditions from the conserved current more generally; see the discussion in Appendix C.

VIII. CONCLUSION

As elaborated in this paper, the study of the continuum limit of a lattice model is a powerful method for extracting the universal, long-distance physics of the system. Different lattice models can share the same continuum, or field-theory, description and, therefore, the same universal behavior. In the context of topological phases, it is instructive to compare our results to those for the Kitaev model for a one-dimensional p -wave superconductor.^{17,18} This is a model proposed for the realization of exotic topological bound states, known as Majorana fermions, with special non-Abelian braiding properties potentially useful for quantum computation. It has recently received great attention as experimental efforts to realize the model in nanowires have produced encouraging results.¹⁸ While the physics of this system is quite different from the one we studied in this paper (e.g., there is no superconductivity in our system), the continuum limits of the two systems are equivalent. This means they have the same topological phase diagram. In particular, the self-adjoint extensions in the continuum and the topological bound states are in one-to-one correspondence to each other. What differs from one continuum model to the other is the specific structure of the quantum fields and, most importantly, the physical properties of the different phases. In fact, it turns out that these two models are not only equivalent in the continuum limit but that their lattice versions are unitarily equivalent (see Appendix D for a proof). So, our lattice and field theory analyses are directly applicable to the study of topological phases of the Kitaev model.

Topological bound states, such as Majorana fermions, are interesting in part for their usefulness in device applications. These applications are often based on dynamical features of the bound states as they are moved around or otherwise manipulated. In such manipulations, the Hamiltonian of the system is changed in time by varying the external parameters of the system. For example, by changing the Zeeman energy μ one can manipulate the position of the bound states. It is worth emphasizing that, when analyzing such schemes in the continuum, it is important to work with a self-adjoint extension of the Hamiltonian since only such an operator can define the time evolution of the system properly.

In summary, through a simple model we have illustrated the importance of self-adjoint extensions in the continuum, or field-theory, description of topological phases of quantum systems with boundaries. In particular, we clarified the physical interpretation of the ex-

tended operators in terms of a conserved local current. These extensions correspond to different physical situations with physically distinct environments outside the system, or equivalently, experimental conditions. We showed that the distinction can persist even after imposing internal symmetries that restrict the choice of the extension. Thus, the notion of bulk-boundary correspondence in a topological phase cannot in general be defined independently from the choice of the extension and the corresponding boundary conditions at the edges of the system.

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Appendix A: Continuum ladder operators

Here we show that the operator $\hat{\Psi}_E$ defined in Eq. (14) is indeed a ladder operator satisfying the commutation relation $[\hat{H}_c, \hat{\Psi}_E^\dagger] = E\hat{\Psi}_E^\dagger$. Using the algebraic relation $[\hat{A}\hat{B}, \hat{C}] = \hat{A}(\hat{B}\hat{C} + \hat{C}\hat{B}) - (\hat{A}\hat{C} + \hat{C}\hat{A})\hat{B}$ and the anticommutation relations (6), we have

$$\begin{aligned}
[\hat{H}_c, \hat{\Psi}_E^\dagger] &= \sum_{s_1 s_2 s'} \int \left[\hat{\Psi}_{s_1}^\dagger(x) \mathcal{H}_{s_1 s_2} \hat{\Psi}_{s_2}(x), \phi_{s'E}(x') \hat{\Psi}_{s'}^\dagger(x') \right] dx dx' \\
&= \sum_{s_1 s_2 s'} \int \mathcal{H}_{s_1 s_2} \phi_{s'E}(x') \hat{\Psi}_{s_1}^\dagger(x) \left[\hat{\Psi}_{s_2}(x) \hat{\Psi}_{s'}^\dagger(x') + \hat{\Psi}_{s'}^\dagger(x') \hat{\Psi}_{s_2}(x) \right] dx dx' \\
&= \sum_{s_1 s_2 s'} \int \mathcal{H}_{s_1 s_2} \phi_{s'E}(x') \hat{\Psi}_{s_1}^\dagger(x) \delta_{s_2 s'} \delta(x - x') dx dx' \\
&= \sum_{s_1 s'} \int \mathcal{H}_{s_1 s'} \phi_{s'E}(x) \hat{\Psi}_{s_1}^\dagger(x) dx \\
&= \sum_{s_1} \int E \phi_{s_1 E}(x) \hat{\Psi}_{s_1}^\dagger(x) dx \\
&= E \hat{\Psi}_E^\dagger,
\end{aligned} \tag{A1}$$

where, in the penultimate line, we also used Eq. (13).

Appendix B: Dense and closed operators

In this appendix we give definitions for densely defined and closed operators. We illustrate these concepts with the momentum operator P .

Definition 1. A subset $\mathcal{S} \subset \mathcal{H}$ is *dense* if for every $\psi \in \mathcal{H}$, there is a sequence $\psi_n \in \mathcal{S}$ that converges to ψ in norm, written simply $\psi_n \rightarrow \psi$, i.e., $\lim_{n \rightarrow \infty} \|\psi_n - \psi\| = 0$.

Definition 2. An operator $A : \mathcal{D}(A) \rightarrow \mathcal{H}$ is a *densely defined operator* if $\mathcal{D}(A)$ is dense in $\mathcal{H}$.

Equivalently, an operator is densely defined if there is an orthonormal basis $\phi_j \in \mathcal{D}(A)$ for the Hilbert space such that any state $\psi \in \mathcal{H}$ can be written uniquely as a superposition of ϕ_j , that is, $\psi = \sum_j a_j \phi_j$, with $a_j = \langle \phi_j | \psi \rangle$. Here, the sum $\sum_j$ is formally understood as a sum over discrete values of j and an integral over continuous values of j .

An operator A must be densely defined in order to uniquely define its adjoint, $A^\dagger$, with the relation $\langle \phi | A^\dagger \psi \rangle = \langle A \phi | \psi \rangle$; when this property holds for all ϕ_j , $A^\dagger \psi$ is uniquely written as $A^\dagger \psi = \sum_j \langle \phi_j | A^\dagger \psi \rangle \phi_j = \sum_j \langle A \phi_j | \psi \rangle \phi_j$.

Definition 3. The operator A is *closed* if for any sequence $\psi_n \in \mathcal{D}(A)$ for which $\psi_n \rightarrow \psi$ and $A\psi_n \rightarrow \phi$, we have $\psi \in \mathcal{D}(A)$ and $\phi = A\psi$.

Let us show that the momentum operator $P = -id/dx$, with

$$\mathcal{D}(P) = \{\psi \in \mathcal{L}^2[0, L] : \psi \text{ is absolutely continuous, } \psi(0) = \psi(L) = 0\}, \quad (\text{B1})$$

is densely defined and closed. First note that there is an orthonormal basis $\phi_j \in \mathcal{D}(P)$, $j \in \mathbb{N}$, given by $\phi_j(x) = \sqrt{2/L} \sin(j\pi x/L)$. For any $\psi \in \mathcal{H}$, the sequence of partial Fourier series ψ_n defined by

$$\psi_n(x) = \sum_{j=1}^n a_j \phi_j(x) \quad \text{with } a_j = \int_0^L \phi_j^*(x) \psi(x) dx \quad (\text{B2})$$

converges to ψ in norm. Thus, $\mathcal{D}(P)$ is dense in $\mathcal{L}^2[0, L]$.

To show that P is closed, we assume a sequence $\chi_n \in \mathcal{D}(P)$ is given so that $\chi_n \rightarrow \psi$ and $P\chi_n \rightarrow \phi$. Since $\mathcal{D}(P)$ is dense, we may write $\psi = \sum_j a_j \phi_j$ and define the partial-sum

sequence $\psi_n \equiv \sum_j^n a_j \phi_j \in \mathcal{D}(P)$ that also converges to ψ , $\psi_n \rightarrow \psi$. By continuity, $P\psi_n \rightarrow \phi$, too. Since $\lim_{n \rightarrow \infty} P\psi_n = -i \lim_{n \rightarrow \infty} \sum_j^n a_j d\phi_j/dx$ exists, $\lim_{n \rightarrow \infty} \psi_n = \psi$ is absolutely continuous and vanishes at the boundaries, $x = 0, L$;¹⁹ therefore, $\psi_n \rightarrow \psi \in \mathcal{D}(P)$. Finally, we show that $P\psi_n \rightarrow P\psi$ and, thus, $\phi = P\psi$:

$$\begin{aligned}
-i \frac{d}{dx} \psi_n(x) &= -i \frac{\pi}{L} \sqrt{\frac{2}{L}} \sum_{j=1}^n j a_j \cos \frac{j\pi x}{L} \\
&= -i \frac{2}{L} \frac{\pi}{L} \sum_{j=1}^n \left[\int_0^L j \sin \frac{j\pi x'}{L} \psi(x') dx' \right] \cos \frac{j\pi x}{L} \\
&= +i \frac{2}{L} \sum_{j=1}^n \left[\int_0^L \left(\frac{d}{dx'} \cos \frac{j\pi x'}{L} \right) \psi(x') dx' \right] \cos \frac{j\pi x}{L} \\
&= -i \int_0^L \left[\frac{2}{L} \sum_{j=1}^n \cos \frac{j\pi x'}{L} \cos \frac{j\pi x}{L} \right] \frac{d}{dx'} \psi(x') dx'; \tag{B3}
\end{aligned}$$

since

$$\lim_{n \rightarrow \infty} \frac{2}{L} \sum_{j=1}^n \cos \frac{j\pi x'}{L} \cos \frac{j\pi x}{L} = \delta(x - x') - \frac{1}{L}, \tag{B4}$$

we conclude that

$$\lim_{n \rightarrow \infty} \|P\psi_n - P\psi\| = 0. \tag{B5}$$

Appendix C: Boundary conditions from conserved current condition

In order to find the boundary conditions from the conserved current of Sec. VII, we note that the current j_A is a local, sesquilinear, Hermitian form of the two states ψ, ϕ , such that $j_A(\psi, \phi) = j_A(\phi, \psi)^*$. A general expression for such a form is $j_A(x) = \nu[\psi(x)]^\dagger J_A \nu[\phi(x)]$, where $\nu[\psi(x)]$ is an n -dimensional vector of linear combinations of the elements of $\psi(x)$ and its derivatives, and $J_A^\dagger = J_A$ is a Hermitian matrix. For example, for the momentum operator $P = -id/dx$, we have $\nu[\psi(x)] = \psi(x)$, $n = 1$, $J_P = 1$.

Exercise 2. For the operator $\Delta = -d^2/dx^2$, show that $\nu[\psi] = (\psi \ d\psi/dx)^\top$, $J_\Delta = \sigma_y$, and $n = 2$.

We can diagonalize $J_A = T^\dagger D T$, with T unitary and D diagonal with real elements. The unitary matrix T can be always chosen such that the diagonal elements of D are sorted in decreasing order. Then $D = (d_+ \oplus d_-)^2$, with $\oplus$ the direct sum, such that $d_\pm = \pm d_\pm^\dagger$ are

Hermitian and anti-Hermitian, respectively, and diagonal; the diagonal elements of $d_{\pm}$ are simply the square roots of, respectively, the positive and negative elements of D .

Exercise 3. For the continuum Hamiltonian, $\mathcal{H} = iv\sigma_y d/dx + m\sigma_z$, show that $\nu[\psi] = \psi$, $J_{\mathcal{H}} = -v\sigma_y$, and $n = 2$. Also, show that $T = e^{i\pi\sigma_x/4} = (\mathbf{1} + i\sigma_x)/\sqrt{2}$,

$$D = v\sigma_z = \begin{pmatrix} \sqrt{v} & 0 \\ 0 & -i\sqrt{v} \end{pmatrix}^2, \quad (\text{C1})$$

and, thus, $d_+ = \sqrt{v}$, $d_- = -i\sqrt{v}$ (up to signs).

Now, writing $J_A = T^\dagger(d_+ \oplus d_-)^2 T = [(d_+ \oplus d_-)^\dagger T]^\dagger [(d_+ \oplus d_-)T]$, we can see that

$$\begin{aligned} j_A &= ((d_+ \oplus -d_-)T\nu[\psi])^\dagger ((d_+ \oplus d_-)T\nu[\phi]) \\ &= \nu_+[\psi]^\dagger \nu_+[\phi] - \nu_-[\psi]^\dagger \nu_-[\phi], \end{aligned} \quad (\text{C2})$$

where we have defined

$$(d_+ \oplus d_-)T\nu[\psi] =: \nu_+[\psi] \oplus \nu_-[\psi]. \quad (\text{C3})$$

We will now consider two different geometries. First, consider a semi-infinite geometry $x \in [0, \infty)$. Then the current condition $j_A(0) = 0$ reads

$$\nu_+[\psi(0)]^\dagger \nu_+[\phi(0)] = \nu_-[\psi(0)]^\dagger \nu_-[\phi(0)]. \quad (\text{C4})$$

If, and only if, d_+ and d_- (and, thus, ν_+ and ν_-) have the same dimension, $n/2$, this condition has a solution in terms of a unitary $\frac{n}{2} \times \frac{n}{2}$ matrix $\bar{u}$ that maps $\nu_+ \mapsto \nu_-$, i.e.,

$$\bar{u}\nu_+[\psi(0)] = \nu_-[\psi(0)]. \quad (\text{C5})$$

Otherwise, A will have no self-adjoint extensions. For example, for the momentum operator $P = -id/dx$, we have $\nu_+[\psi(0)] = \psi(0)$ and ν_- does not exist. Therefore, P cannot have a self-adjoint extension on the semi-infinite line. For the continuum Hamiltonian, $\mathcal{H}$, we have $\nu_{\pm}[\psi(0)] = \sqrt{v/2}[\psi_{\uparrow}(0) \pm i\psi_{\downarrow}(0)]$. Therefore, the self-adjoint extensions of $\mathcal{H}$ are characterized by a $U(1)$ phase $\bar{u} = e^{i\theta}$; after simple algebra, Eq. (C5) reads $\psi_{\uparrow}(0) = \cot(\theta/2)\psi_{\downarrow}(0)$, which is the same as Eq. (30) with $\delta = \pi/2$.

Now consider a finite geometry $x \in [0, L]$. The current condition, $j_A(L) = j_A(0)$, yields

$$\nu_+[\psi(0)]^\dagger \nu_+[\phi(0)] + \nu_-[\psi(L)]^\dagger \nu_-[\phi(L)] = \nu_-[\psi(0)]^\dagger \nu_-[\phi(0)] + \nu_+[\psi(L)]^\dagger \nu_+[\phi(L)] \quad (\text{C6})$$

$$\Rightarrow \quad \varsigma_+[\psi]^\dagger \varsigma_+[\phi] = \varsigma_-[\psi]^\dagger \varsigma_-[\phi], \quad (\text{C7})$$

where

$$\varsigma_{\pm}[\psi] \equiv \nu_{\pm}[\psi(0)] \oplus \nu_{\mp}[\psi(L)]. \quad (\text{C8})$$

Thus, the state-independent boundary conditions are given by a norm-preserving, $n \times n$ unitary matrix u that maps $\varsigma_+ \mapsto \varsigma_-$, i.e.,

$$u \varsigma_+[\psi] = \varsigma_-[\psi]. \quad (\text{C9})$$

In the case of the momentum operator $P = -id/dx$ we see immediately that now $\varsigma_+[\psi] = \psi(0)$ and $\varsigma_-[\psi] = \psi(L)$. Thus, the self-adjoint extensions of P are characterized by a $U(1)$ phase as $\psi(L) = e^{i\beta}\psi(0)$. In the case of the continuum formulation of the topological wire, we find

$$\varsigma_+ = -i\sqrt{\frac{v}{2}} \begin{pmatrix} -\psi_{\downarrow}(0) + i\psi_{\uparrow}(0) \\ \psi_{\downarrow}(L) + i\psi_{\uparrow}(L) \end{pmatrix}, \quad \varsigma_- = -i\sqrt{\frac{v}{2}} \begin{pmatrix} \psi_{\downarrow}(0) + i\psi_{\uparrow}(0) \\ -\psi_{\downarrow}(L) + i\psi_{\uparrow}(L) \end{pmatrix}. \quad (\text{C10})$$

Thus, from Eq. (C9) we find the boundary conditions in Eq. (41).

Appendix D: Unitary equivalence between a ferromagnetic insulator and a p -wave superconductor

Here we prove that the ferromagnetic insulator model in Eq. (1) is unitarily equivalent to a p -wave superconductor. More precisely, the superconducting model represents two uncoupled Kitaev chains¹⁷ that share identical energy spectra.

Define the local canonical map

$$c_{\uparrow r}^{\dagger} = \frac{f_r^{\dagger} + ig_r^{\dagger}}{\sqrt{2}}, \quad c_{\downarrow r}^{\dagger} = \frac{f_r + ig_r}{\sqrt{2}}, \quad (\text{D1})$$

in terms of two new sets of fermion creation and annihilation operators, $f_r^{\dagger}$ and f_r , and $g_r^{\dagger}$ and g_r . It is straightforward to show that the f operators commute with the g operators and each pair separately satisfies fermionic anti-commutation relations $\{f_r, f_{r'}^{\dagger}\} = \{g_r, g_{r'}^{\dagger}\} = \delta_{rr'}$, etc. Using these relation and after some algebraic manipulations, the Hamiltonian $\hat{H}$ of Eq. (1) becomes

$$\hat{H} + \mu N = \sum_{r=1}^N \left[\frac{\mu}{2} (f_r^{\dagger} f_r + g_r^{\dagger} g_r) + w (f_r^{\dagger} f_{r+1} + g_r^{\dagger} g_{r+1}) + \frac{\lambda}{2} (f_r^{\dagger} f_{r+1}^{\dagger} + g_r^{\dagger} g_{r+1}^{\dagger}) \right] + \text{h.c.}, \quad (\text{D2})$$

where, as before, the sum is understood not to contain any terms with the index $N + 1$. Thus, the Hamiltonian is mapped to two copies of the Kitaev model for f and g separately.

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- ⁸ This is analogous to a nontrivial knot that cannot be undone without cutting. Locally, the knot has the same geometry as a loop; however, there is a global twist that distinguishes it from a simple loop. Technically, the twist in the wavefunction is expressed in terms of Berry’s phases; see Ref. 3.
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- ¹² The transpose operation T is defined as

$$(\phi_1(x) \ \phi_2(x))^{\mathsf{T}} = \begin{pmatrix} \phi_1(x) \\ \phi_2(x) \end{pmatrix}.$$

- ¹³ M. Reed and B. Simon, *Methods of Modern Mathematical Physics I: Functional Analysis* (Academic Press, New York, 1972).
- ¹⁴ An absolutely continuous function ϕ on an interval $I \subset \mathbb{R}$ is defined as one for which there exists an integrable function ϕ' that satisfies $\int_a^b \phi'(x) dx = \phi(b) - \phi(a)$ for any $a, b \in I$ (see, for example, p. 22 in Ref. 6). Then, according to the fundamental theorem of calculus, $\phi' = d\phi/dx$ almost everywhere in I .
- ¹⁵ Note that the states $\psi_{\pm}$ do not belong to $\mathcal{D}(P)$ since $\psi_{\pm}(0), \psi_{\pm}(L) \neq 0$.
- ¹⁶ M. Reed and B. Simon, *Methods of Modern Mathematical Physics II: Fourier analysis, Self-adjointness* (Academic Press, New York, 1975).
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